第五章
弯曲应力

Chapter5 Stresses in beams

内容

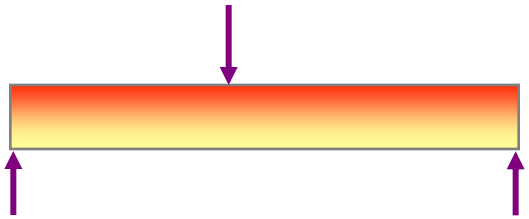
§ 5-1 纯弯曲 (pure bending)

§ 5-2 纯弯曲时的正应力 (Normal stresses in pure beams)

§ 5-3 横力弯曲时的正应力(Normal stresses in transverse bending)

§ 5-4 弯曲切应力 (Shear stresses in beams)

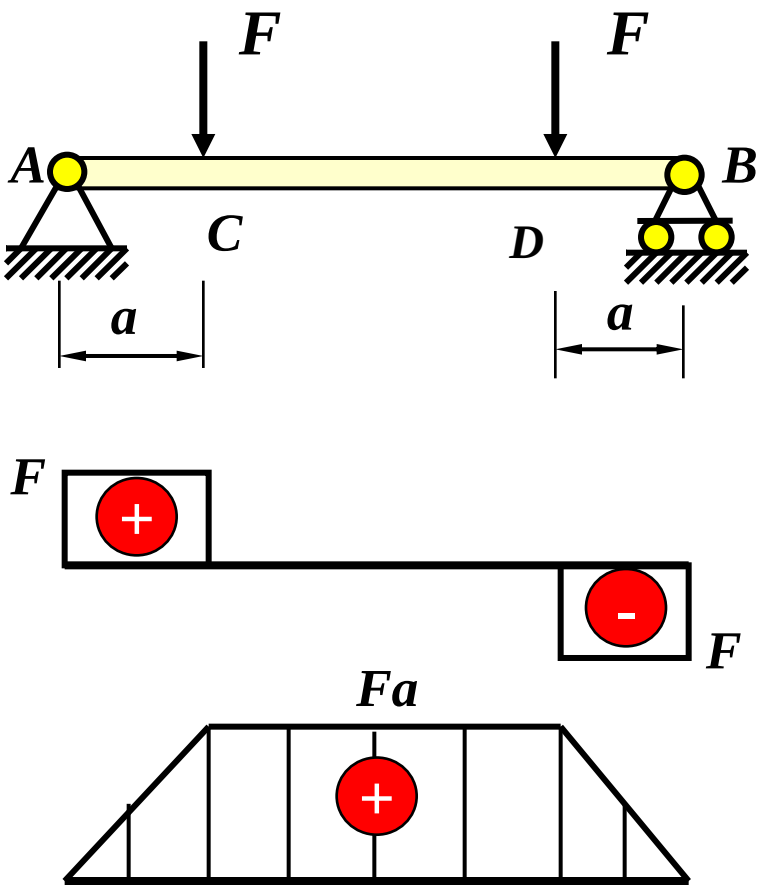
§ 5-5 提高梁强度的主要措施（ Measuresto strengthen the strength of beams)

变形形式	构件	内力	应力
拉（压）		轴力F_N	$\sigma = \frac{F_N}{A}$
扭转		扭矩T	$\tau = \frac{T\rho}{I_\rho}$
弯曲		弯矩M 剪力F_S	?

§5-1 纯弯曲 (Pure bending)

一、纯弯曲（Pure bending)

若梁在某段内各横截面的弯矩为常量，剪力为零，则该段梁的弯曲就称为**纯弯曲**。既有弯矩又有剪力该段梁称为**横力弯曲**（剪切弯曲）。



简支梁CD段是纯弯曲，AC、DB为横力弯曲。

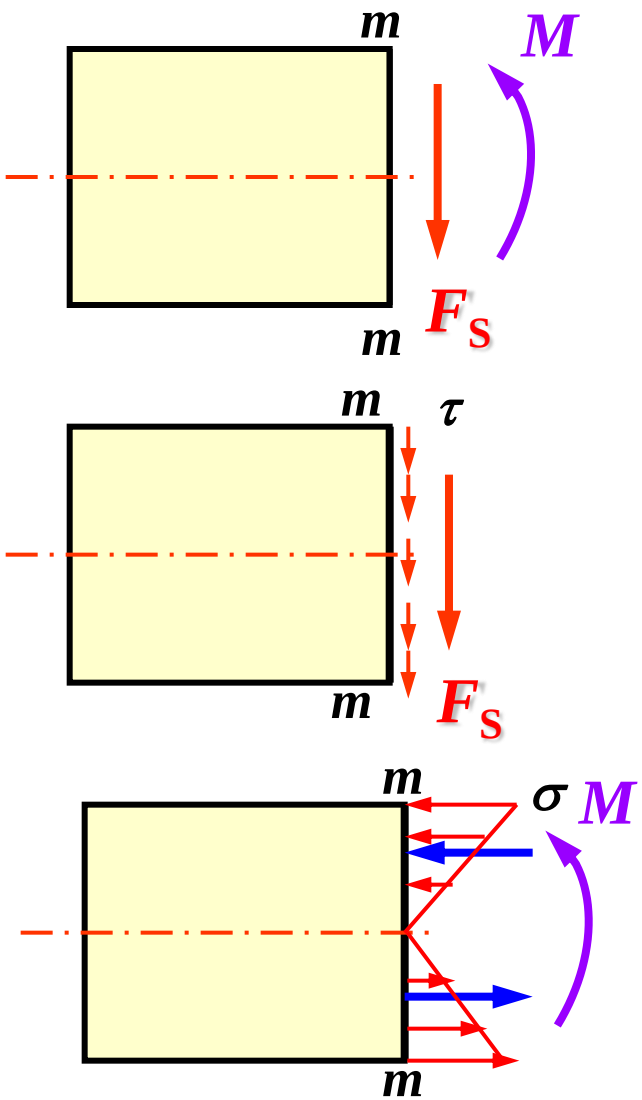
二、弯曲构件横截面上的应力 (Stresses in flexural members)

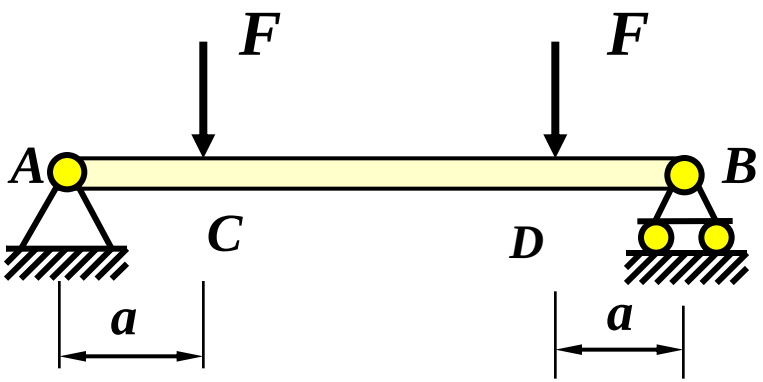
剪力F_S是平行于横截面的内力合力，所以切应力的合力是剪力。

弯矩M是垂直于横截面的内力系的合力偶矩，所以正应力的合力是弯矩。

内力 $\begin{cases} \text{剪力}F_S \longrightarrow \text{切应力}\tau \\ \text{弯矩}M \longrightarrow \text{正应力}\sigma \end{cases}$

所以，在梁的横截面上一般既有正应力，又有切应力。



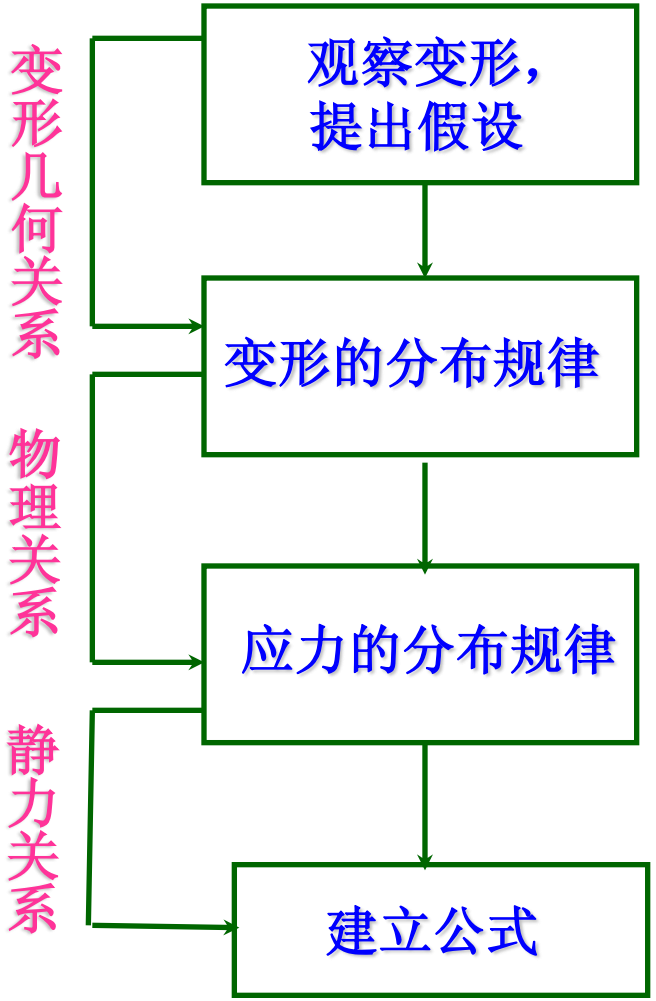


纯弯曲梁 → 有M → 正应力σ

横力弯曲 → 有F_S、M → 正应力σ、切应力τ

§5-2 纯弯曲时的正应力

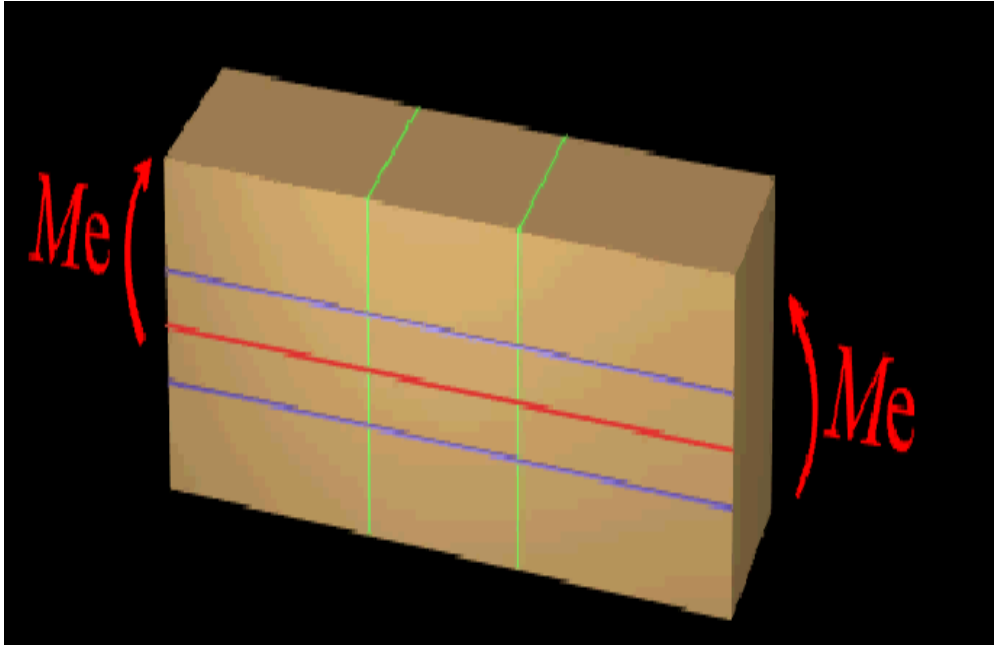
(Normal stresses in pure beams)

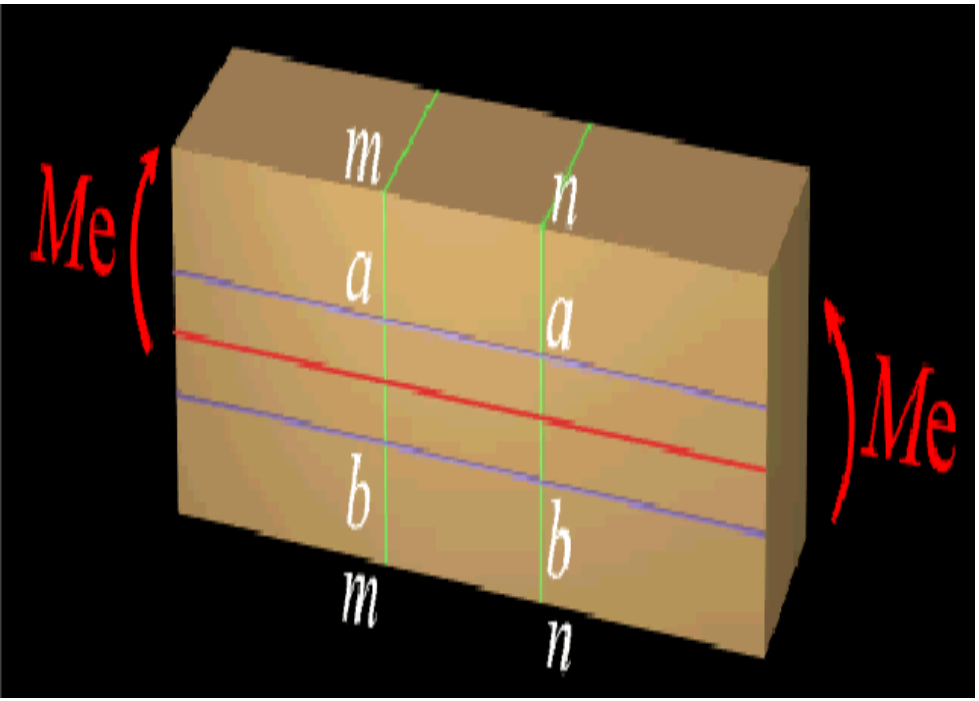


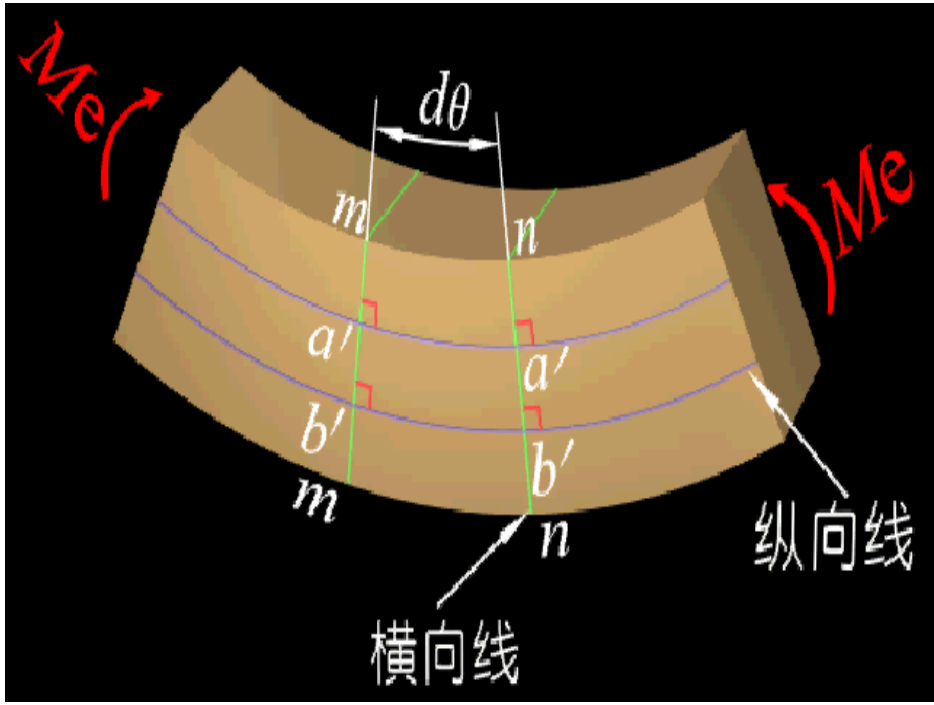
一、实验（ Experiment）

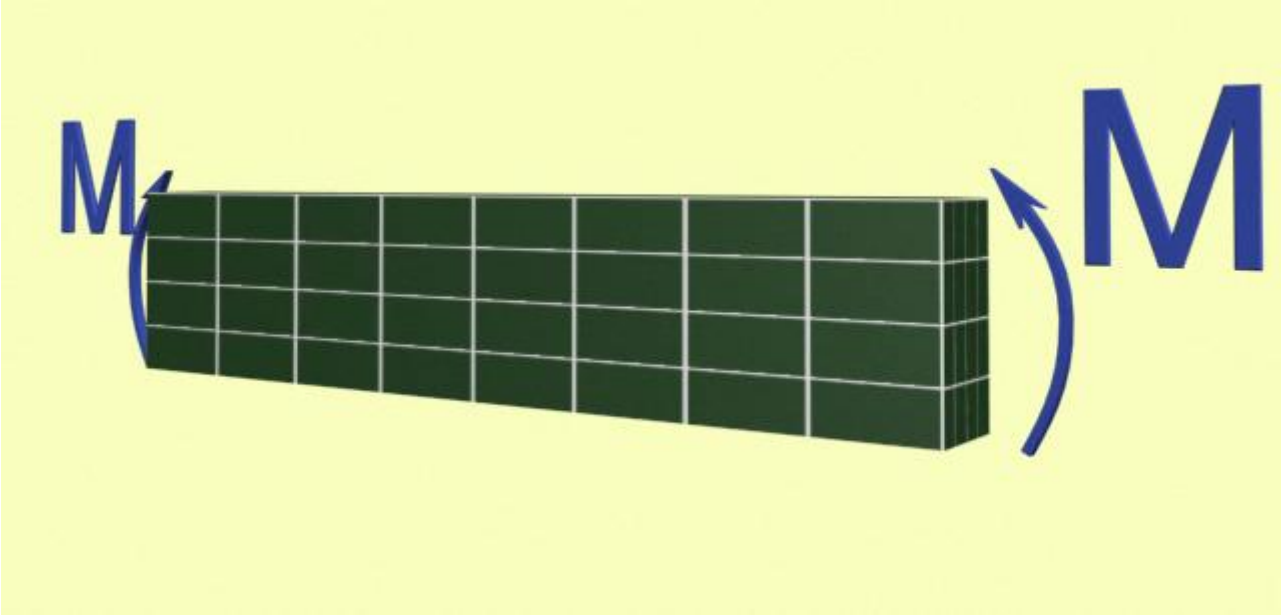
1、变形现象

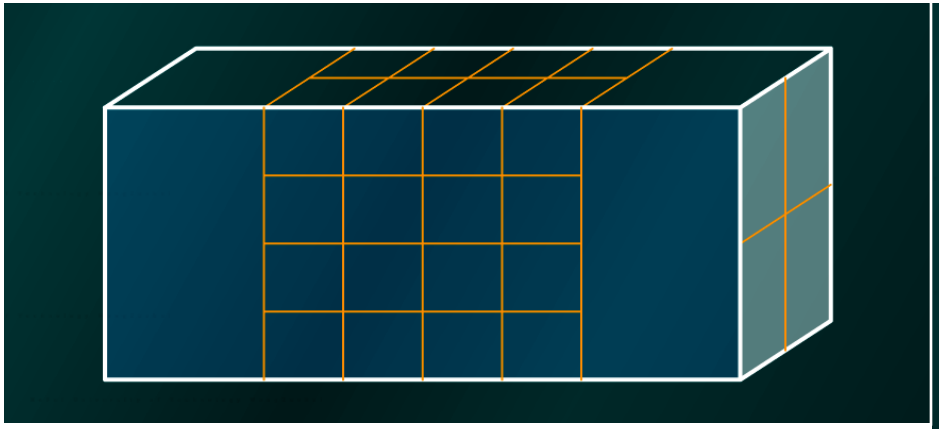
（Deformation phenomenon)

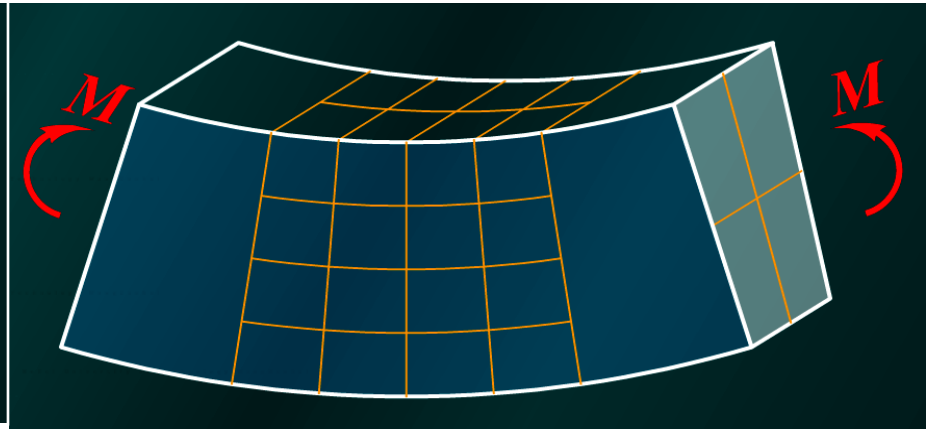


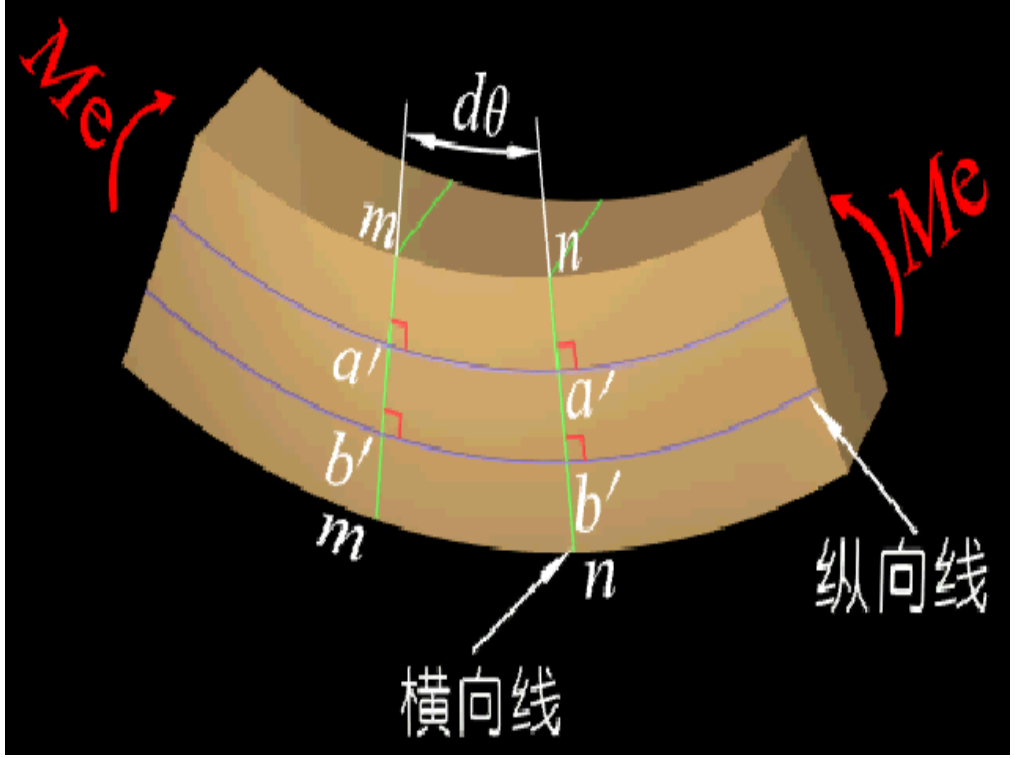


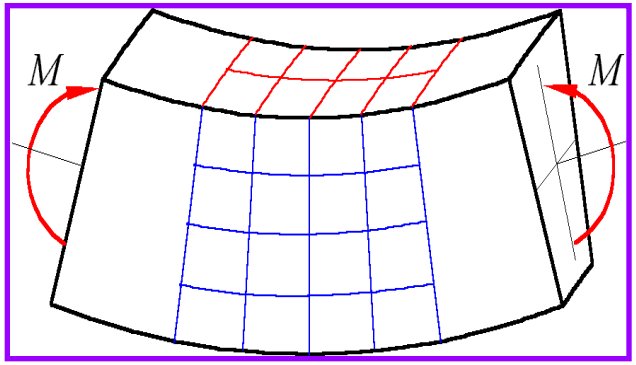



M
M




M
M



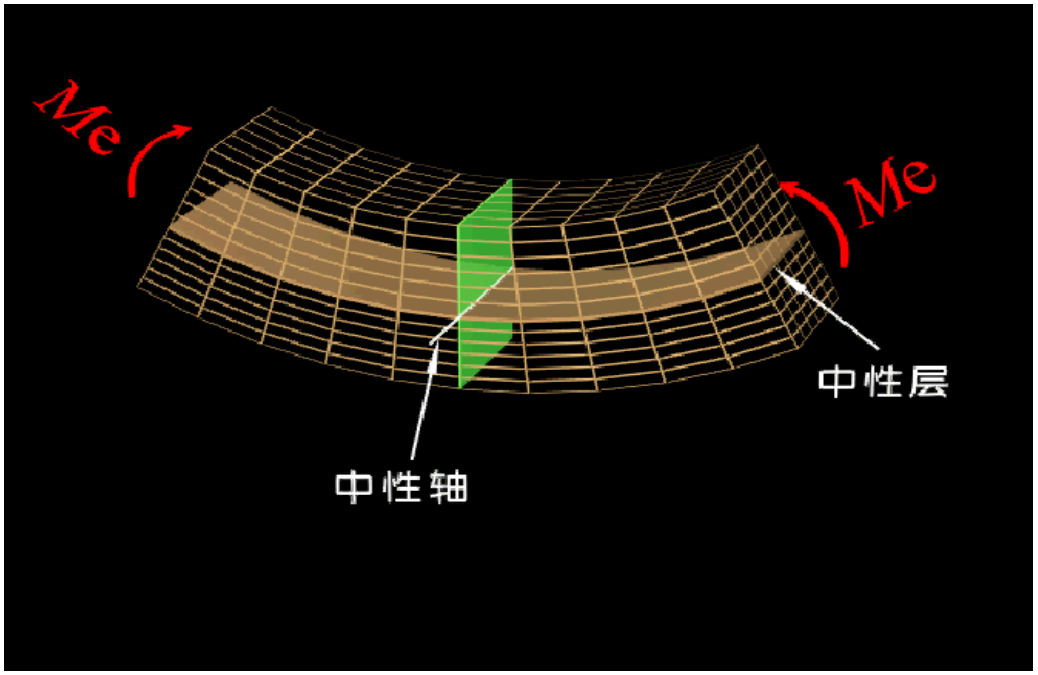


纵向线： 各纵向线段弯成弧线，且靠近顶端的纵向线缩短，靠近底端的纵向线段伸长。

横向线： 各横向线仍保持为直线，相对转过了一个角度，仍与变形后的纵向弧线垂直。

横截面： 纵线伸长区，截面宽度减小 纵线缩短区，截面宽度增大。

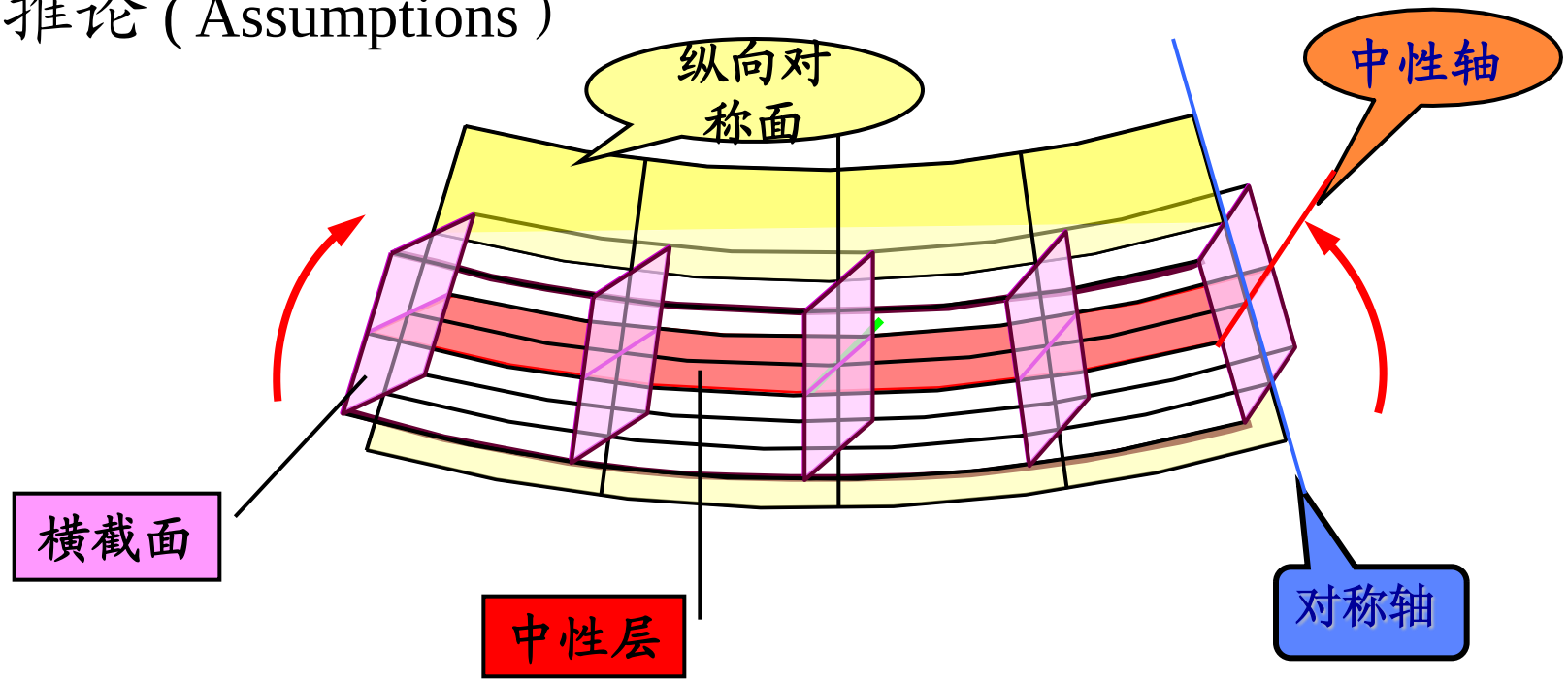
2、提出假设（Assumptions）



（a）**平面假设**：变形前为平面的横截面变形后仍保持为平面且垂直于变形后的梁轴线;

（b）**单向受力假设**：纵向纤维不相互挤压，只受单向拉压。

3、推论（Assumptions）



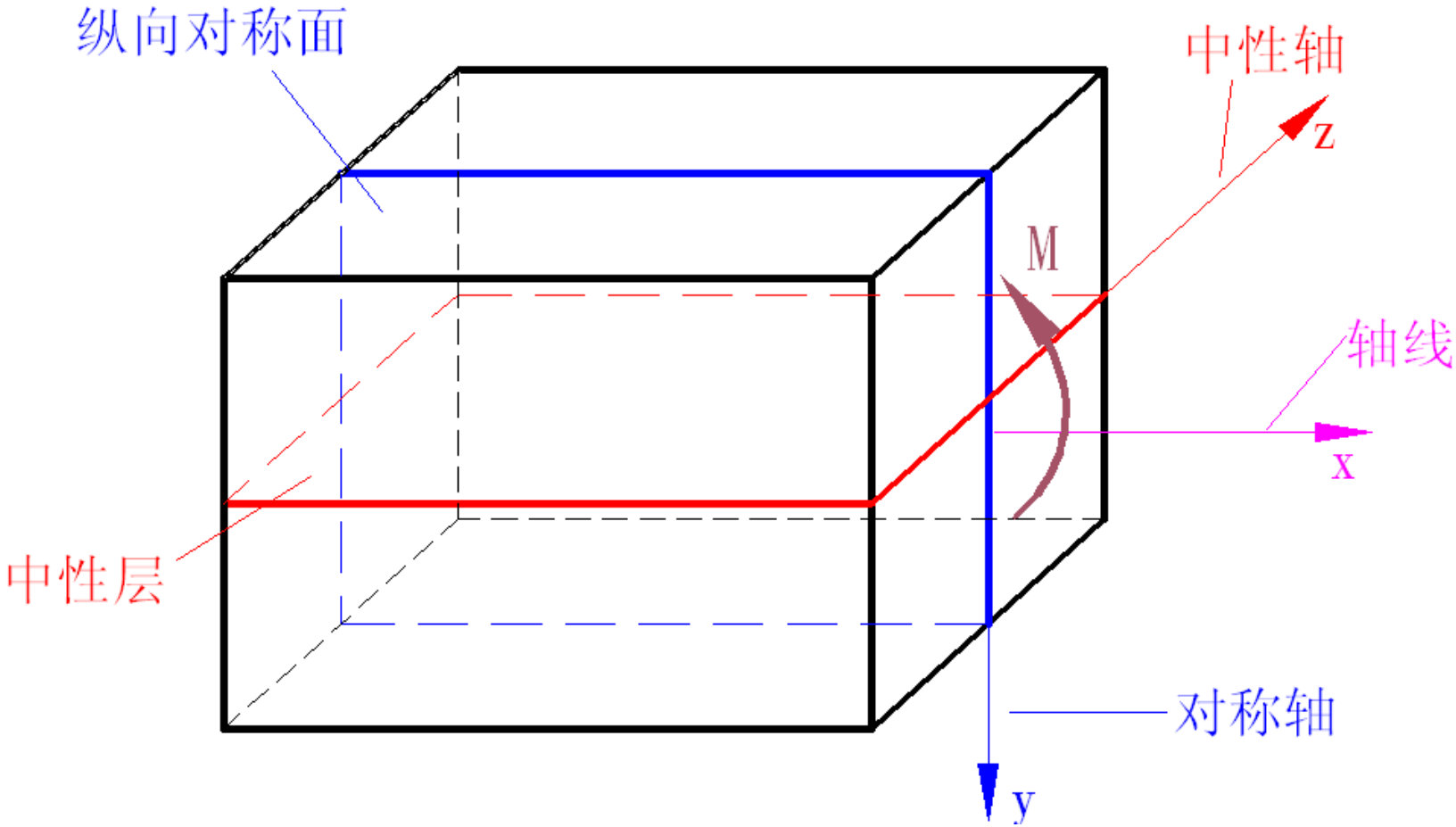
凹入一侧纤维缩短
突出一侧纤维伸长
连续性假设
平面假设

⇒ 必有一层变形前后长度不变的纤维—中性层

中性轴： 中间层与横截面的交线。中性轴⊥对称轴。

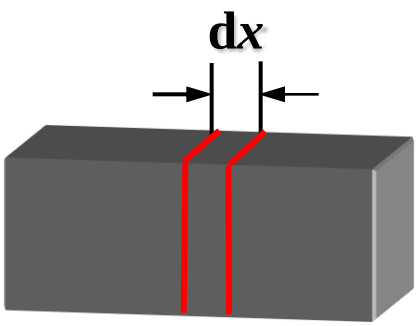
横截面象刚性平面一样绕中性轴**相对转动。**

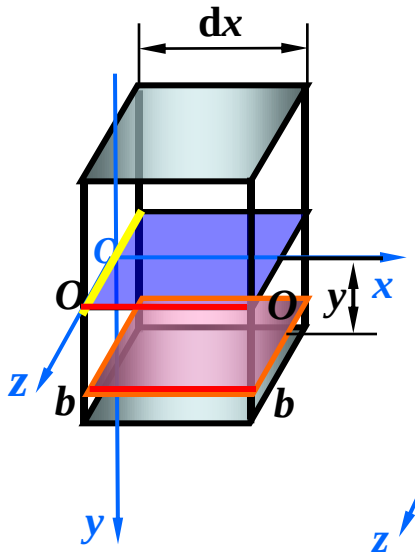
4、建立坐标系

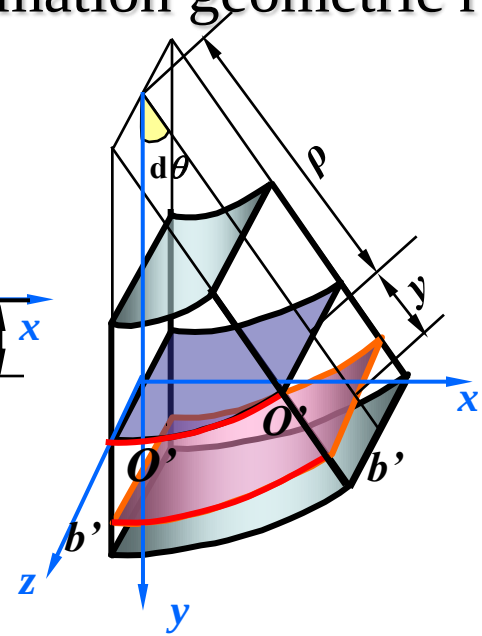


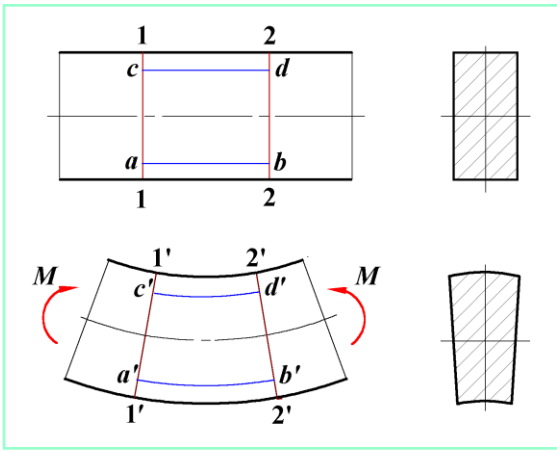
坐标系
- x轴----轴线
- y轴----对称轴
- z轴----中性轴 ?

二、变形几何关系（ Deformation geometric relation ）









$$\left.\begin{aligned} \overset{\frown}{b'b'} &= (\rho + y)\mathrm{d}\theta \\ \overline{bb} = \mathrm{d}x = \overline{OO} &= \overset{\frown}{O'O'} = \rho\,\mathrm{d}\theta \end{aligned}\right\} \longrightarrow \boxed{\varepsilon = \frac{(\rho + y)\mathrm{d}\theta - \rho\,\mathrm{d}\theta}{\rho\,\mathrm{d}\theta} = \frac{y}{\rho}}$$

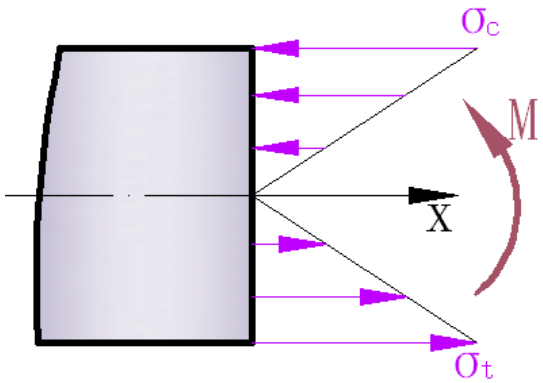
应变分布规律：直梁纯弯曲时纵向纤维的应变

与它到中性层的距离成正比。

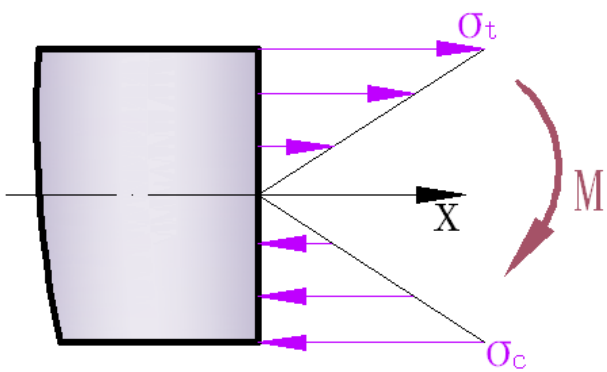
三、物理关系 (Physical relationship)

由单向受力假设知各纵向纤维象简单拉压一样，所以满足**胡克定理**。

$$\left.\begin{aligned}\sigma &= E\varepsilon \\ \varepsilon &= \frac{y}{\rho}\end{aligned}\right\} \Longrightarrow \boxed{\sigma = E\frac{y}{\rho}}$$



应力分布规律：直梁纯弯曲时横截面上任意一点的正应力，与它到中性轴的距离成**正比**。



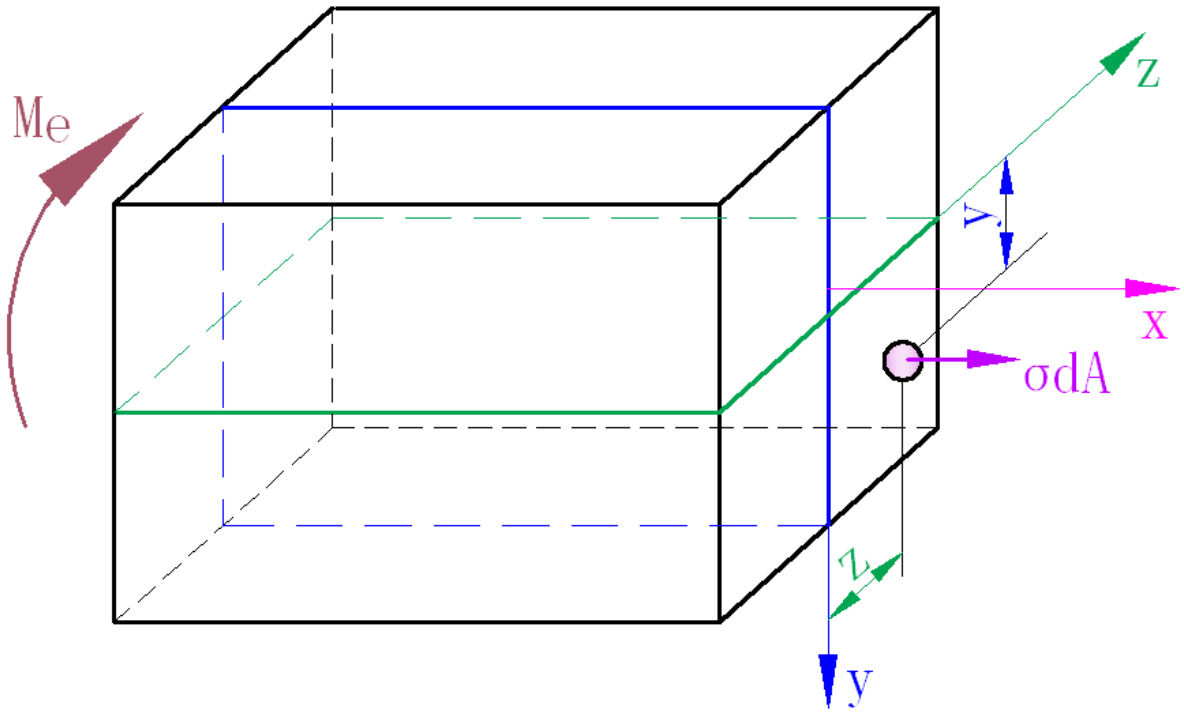
待解决问题

- 中性轴的位置
- 中性层的曲率半径ρ

?

四、静力关系 (Static relationship)

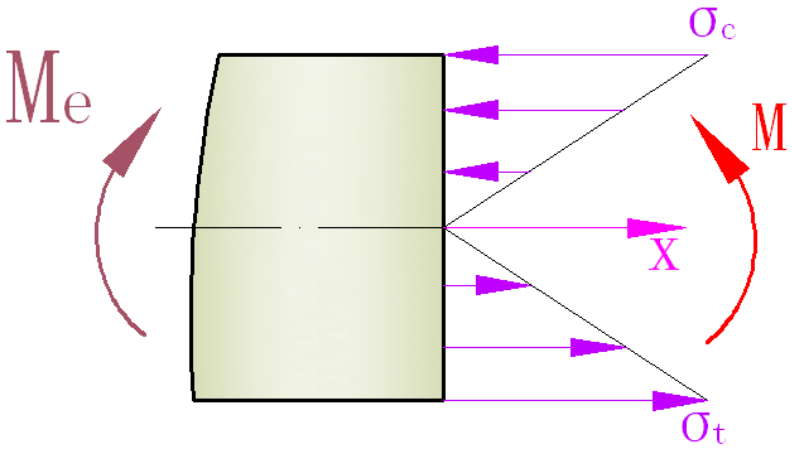
横截面上内力系为垂直于横截面的空间平行力系，这一力系简化得到三个内力分量。



$$F_N = \int_A dF_N = \int_A \sigma dA = 0 \quad (1)$$

$$M_{iy} = \int_A dM_y = \int_A z\sigma dA = 0 \quad (2)$$

$$M_{iz} = \int_A dM_z = \int_A y\sigma dA = M \quad (3)$$



将应力表达式代入（1）式，得

$$F_N = \int_A E\frac{y}{\rho}\mathrm{d}A = 0 \Rightarrow \frac{E}{\rho}\int_A y\mathrm{d}A = 0 \Rightarrow S_z = \int_A y\mathrm{d}A = 0$$

$\Rightarrow$ 中性轴通过横截面形心

将应力表达式代入（2）式，得

$$M_{iy} = \int_A zE\frac{y}{\rho}\mathrm{d}A = 0 \Rightarrow \frac{E}{\rho}\int_A yz\mathrm{d}A = 0 \Rightarrow I_{yz} = \int_A yz\mathrm{d}A = 0$$

$\Rightarrow$ 自然满足

将应力表达式代入(3)式，得

$$M_{iz} = \int_A yE\frac{y}{\rho}\mathrm{d}A = M \Rightarrow \frac{E}{\rho}\int_A y^2\mathrm{d}A = M \Rightarrow \frac{E}{\rho}I_z = M$$

$$\Rightarrow \frac{1}{\rho} = \frac{M}{EI_z}$$

将 $\frac{1}{\rho}=\frac{M}{EI_z}$ 代入 $\boldsymbol{\sigma = E\frac{y}{\rho}}$

得到纯弯曲时横截面上正应力的计算公式:

$$\boxed{\sigma=\frac{\mathrm{My}}{\mathrm{I_z}}}$$

M为梁横截面上的弯矩;

y为梁横截面上任意一点到中性轴的距离;

I_z为梁横截面对中性轴的惯性矩。

五、公式的应用范围

(The applicable range of the flexure formula)

（1）公式中没有和材料有关的参数弹性模量E，说明正应力σ和材料无关，但公式的推导中用到了胡克定律，该公式适用于弹性变形阶段。

（2）梁只要有一个纵向对称面，横截面惯性积$I_{yz}=0$，并载荷作用在此平面内，都可应用该公式，和具体截面形状无关。

（3）应用公式时，一般将M、y以绝对值代入。根据梁变形的情况直接判断σ的正负号。以中性轴为界，梁变形后凸出边的应力为拉应力(σ为正号)，凹入边的应力为压应力（σ为负号）。

（4）用于直梁。

§5-3 横力弯曲时的正应力

(Normal stresses of the beam in nonuniform bending)

一、横力弯曲(Nonuniform bending)

横力弯曲时，梁的横截面上既又弯矩又有剪力，横向力引起与中性层平行的纵截面的挤压应力，纯弯曲时所作的平面假设和单向受力假设都不成立。

虽然横力弯曲与纯弯曲存在这些差异，但进一步的分析表明，工程中常用的梁当梁跨度L与高度h之比大于5（即为细长梁）时，弹性力学指出：上述公式近似成立。计算横力弯曲时横截面上的正应力仍用纯弯曲时的正应力计算公式。

二、强度条件（Strength condition)

1、数学表达式
（Mathematical formula)

$$\sigma = \frac{My}{I_z}$$

最大正应力发生在横截面上离中性轴最远的点处。

$$\boldsymbol{\sigma_{max} = \frac{M\,y_{max}}{I_z}}$$

引用记号 $W = \dfrac{I_z}{y_{max}}$

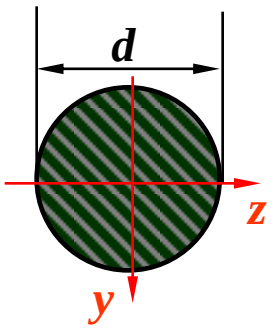
W为抗弯截面系数，单位：m^3，mm^3

则公式改写为 $\sigma_{max} = \dfrac{M}{W}$ $\sigma_{max} = \dfrac{M}{W} \le [\sigma]$

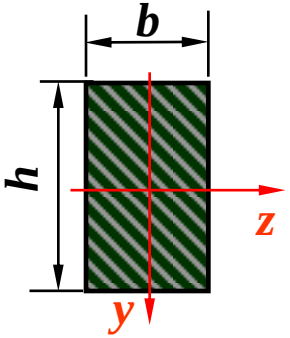
2、抗弯截面系数

（1）当中性轴为对称轴时

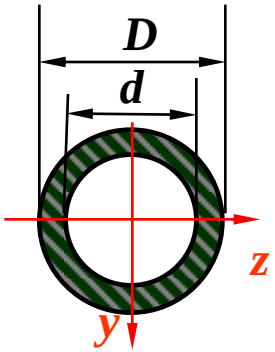
实心圆截面 $$W=\frac{I_z}{d/2}=\frac{\pi d^4/64}{d/2}=\frac{\pi d^3}{32}$$



矩形截面 $$W=\frac{I_z}{h/2}=\frac{bh^3/12}{h/2}=\frac{bh^2}{6}$$



空心圆截面 $$W=\frac{\pi D^3}{32}(1-\alpha^4) \qquad \alpha=\frac{d}{D}$$



弯曲变形和扭转变形强度计算公式发比较。

$$\sigma = \frac{My}{I_z}$$

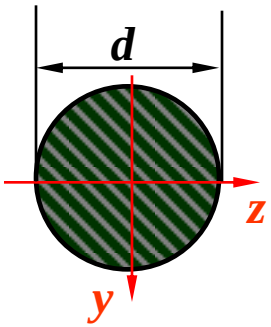
$$\sigma_{max} = \frac{M}{W}$$

$$\tau = \frac{T\rho}{I_\rho}$$

$$\tau_{max} = \frac{T}{W_t}$$

$$I_z = \frac{\pi d^4}{64}$$

$$W = \frac{\pi d^3}{32}$$

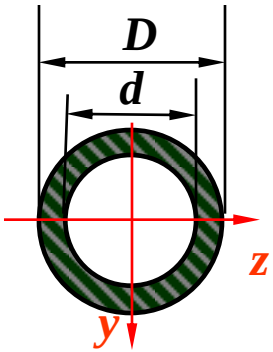


$$I_\rho = \frac{\pi d^4}{32}$$

$$W_t = \frac{\pi d^3}{16}$$

$$I_z = \frac{\pi(D^4 - d^4)}{64}$$

$$W = \frac{\pi(D^4 - d^4)}{32D}$$

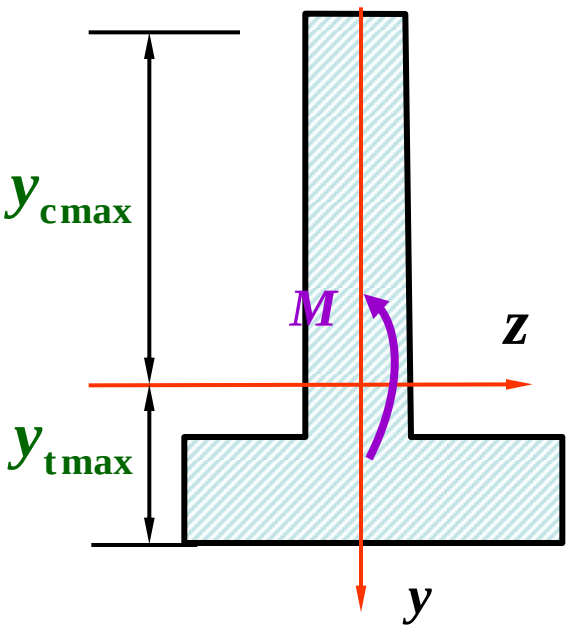


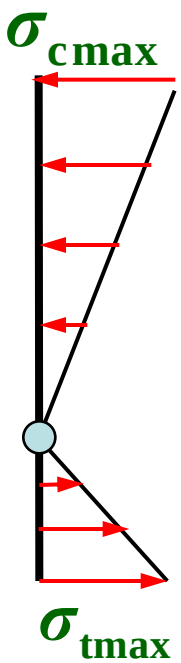
$$I_\rho = \frac{\pi(D^4 - d^4)}{32}$$

$$W_t = \frac{\pi(D^4 - d^4)}{16D}$$

（2）对于中性轴不是对称轴的横截面

应分别以横截面上受拉和受压部分距中性轴最远的距离y_{tmax}和y_{cmax}直接代入公式。





$$\sigma = \frac{My}{I_z}$$

$$\sigma_{tmax} = \frac{My_{tmax}}{I_z}$$

$$\sigma_{cmax} = \frac{My_{cmax}}{I_z}$$

（3）轧制型钢（工字钢、槽钢等）的抗弯截面系数W从型钢表（见附录Ⅱ）中查得。

2、强度条件的应用(Application of strength condition)

（1） 强度校核 $$\frac{M_{\max}}{W} \le [\sigma]$$

（2）设计截面 $$W \ge \frac{M_{\max}}{[\sigma]}$$

（3）确定许可载荷 $$M_{\max} \le W[\sigma]$$

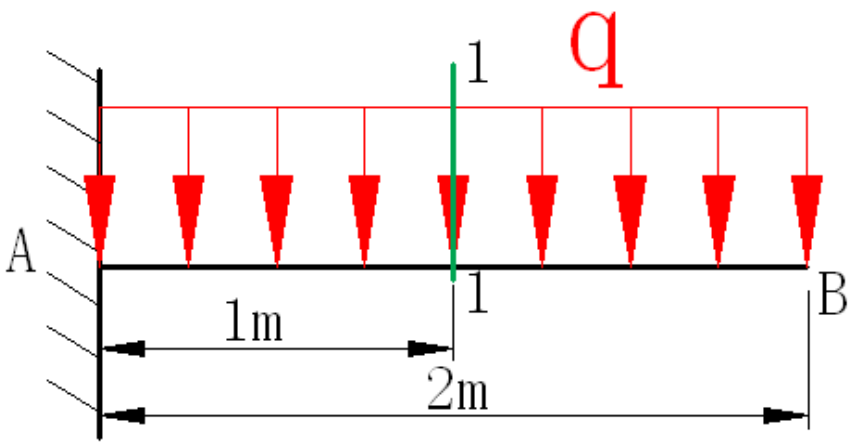
对于铸铁等脆性材料制成的梁，由于材料的 $[\sigma_t] \ne [\sigma_c]$，且梁横截面的中性轴一般也不是对称轴，所以梁的 $\sigma_{t\max} \ne \sigma_{c\max}$，两者有时并不发生在同一横截面上。

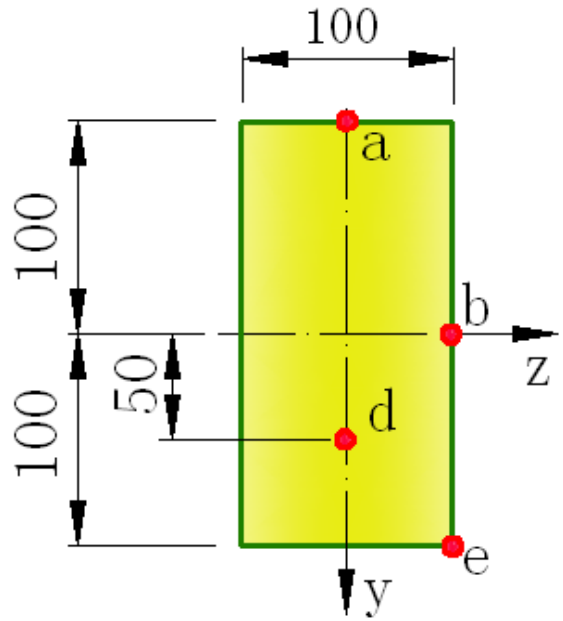
$$\sigma_{t\max} \le [\sigma_t]$$

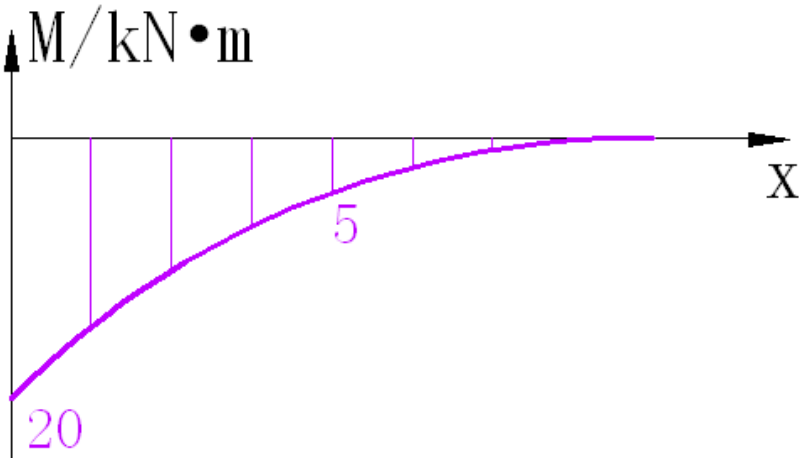
$$\sigma_{c\max} \le [\sigma_c]$$

例题 5-1

q=10kN/m，尺寸如图。求：（1）1-1截面上a、b、d、e四点的正应力；（2）全梁σ_{max}。







$\sigma_a = 7.5\text{MPa}$（拉）

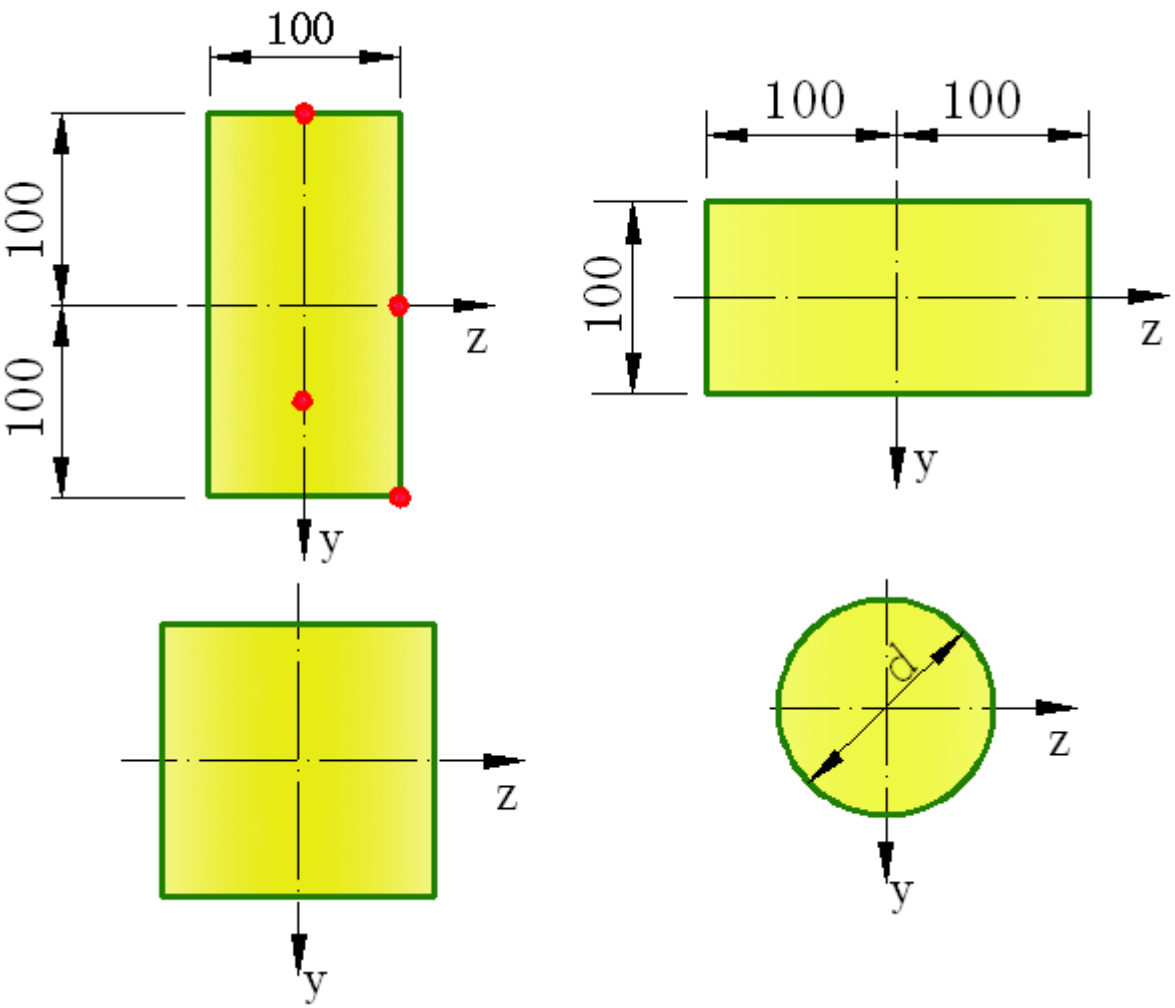
$\sigma_b = 0$

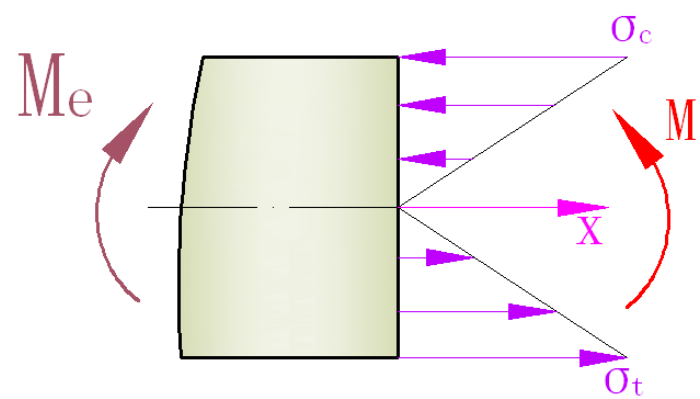
$\sigma_d = 3.75\text{MPa}$（压）

$\sigma_e = 7.5\text{MPa}$（压）

$\sigma_{max} = 30\text{MPa}$

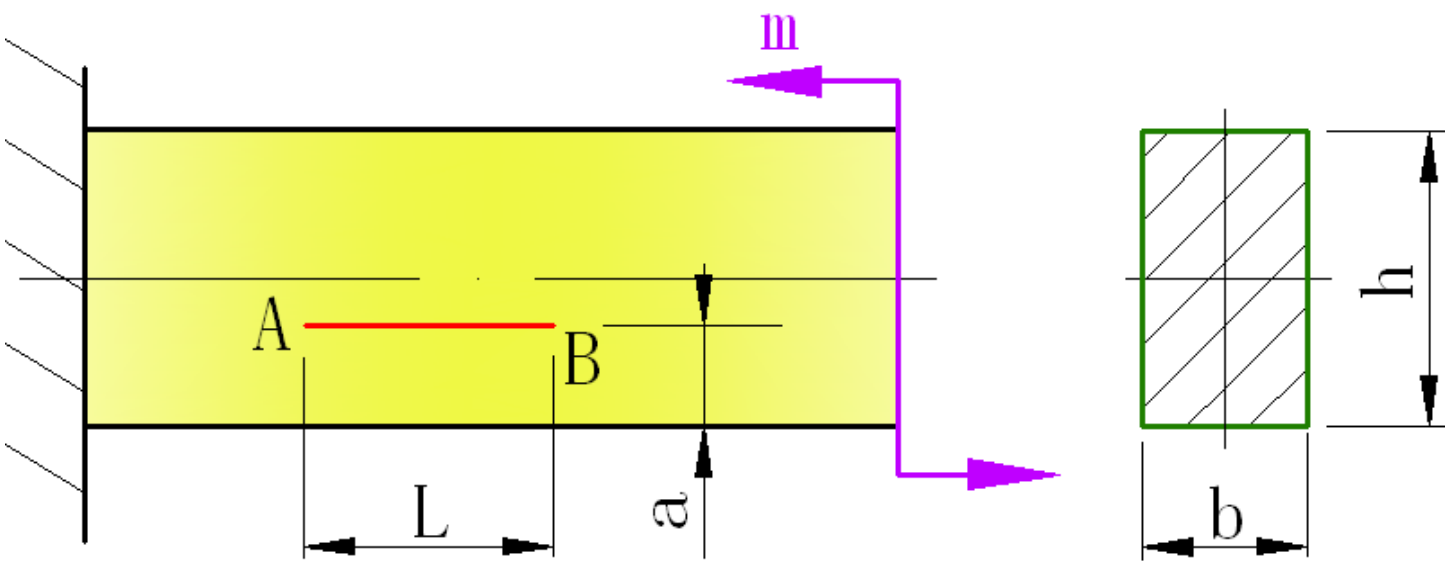
在同等条件下，比较各截面上的应力。





例题 5-2

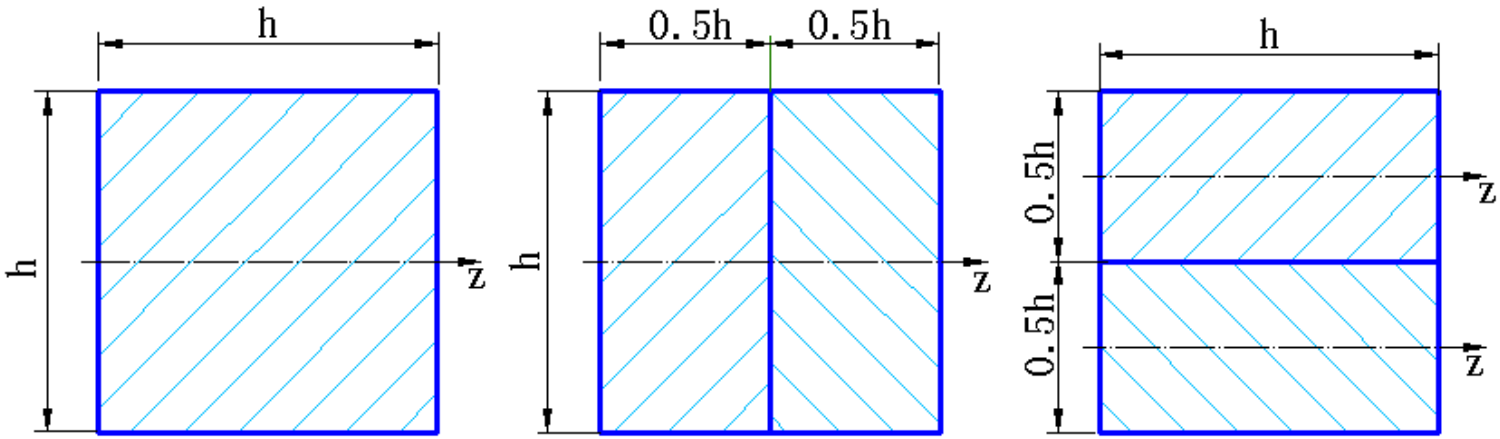
在矩形截面梁的表面刻有标记，已知b、h、a、L，材料的弹性模量为E，AB∥轴线，AB的伸长量为ΔL。求：m=？。



$$m = \frac{E \cdot \frac{\Delta L}{L} \cdot \frac{bh^3}{12}}{\frac{h}{2} - a}$$

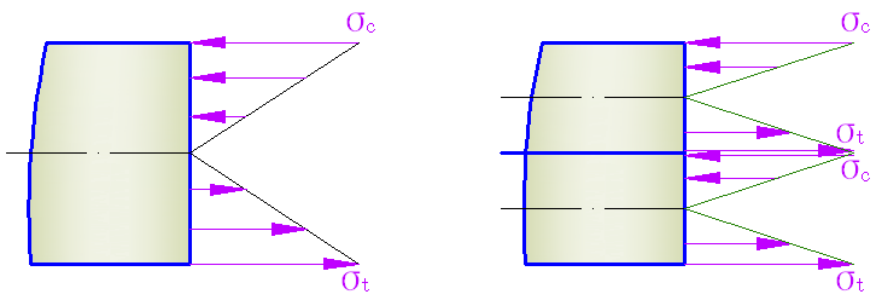
例题 5- 5

承受相同弯矩M_z的三根直梁，其截面组成方式如图所示。图（a）的截面为一整体；图（b）的截面由两矩形截面并列而成（未粘接）；图（c）的截面有两矩形截面上下叠合而成（未粘接）。三根梁中的最大正应力分别是多少？



（a）　　（b）　　（c）

$$\sigma_{max}(a)=\frac{M_z}{\frac{h^3}{6}}=\frac{6M_z}{h^3}$$



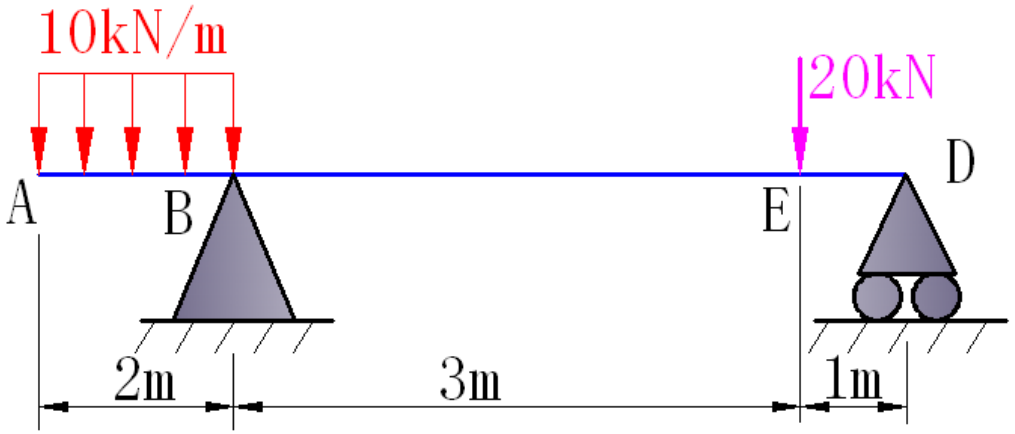
$$\sigma_{max}(b)=\frac{\frac{M_z}{2}}{\frac{\frac{h}{2}.h^2}{6}}\cdot=\frac{6M_z}{h^3}$$

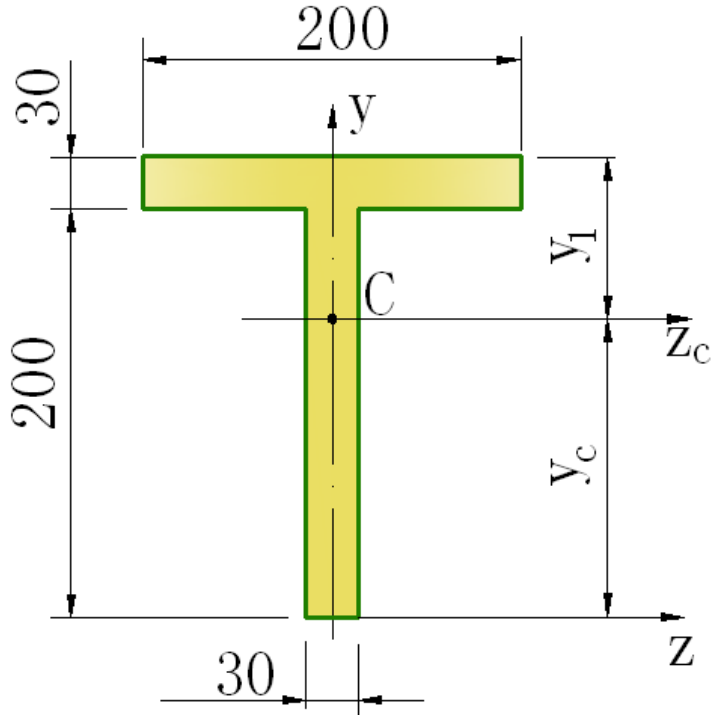
$$\sigma_{max}(c)=\frac{\frac{M_z}{2}}{\frac{h\left(\frac{h}{2}\right)^2}{6}}\cdot=\frac{12M_z}{h^3}$$

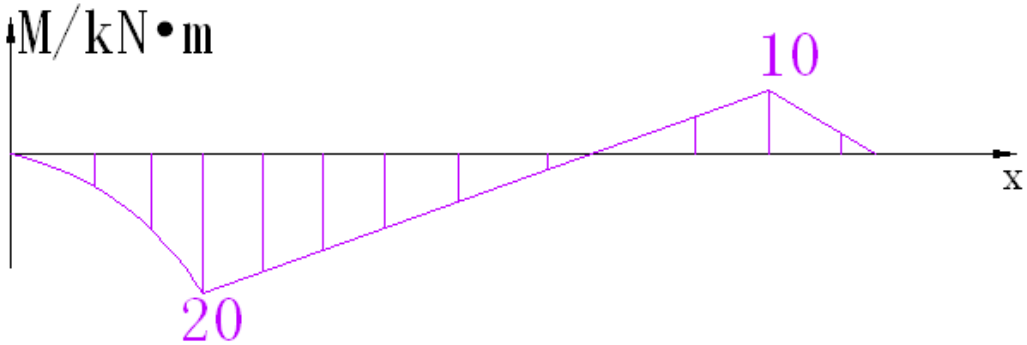
例题 5-3

铸铁梁，载荷和尺寸如图所示。许用拉应力[σt]=40MPa]，许用压应力[σ_c]=160MPa。

求：（1）校核梁的强度。（2）若载荷不变，将梁截面由“T”变为“⊥”是否合理？







$F_B = 30kN$（↑），$F_D = 10kN$（↑）

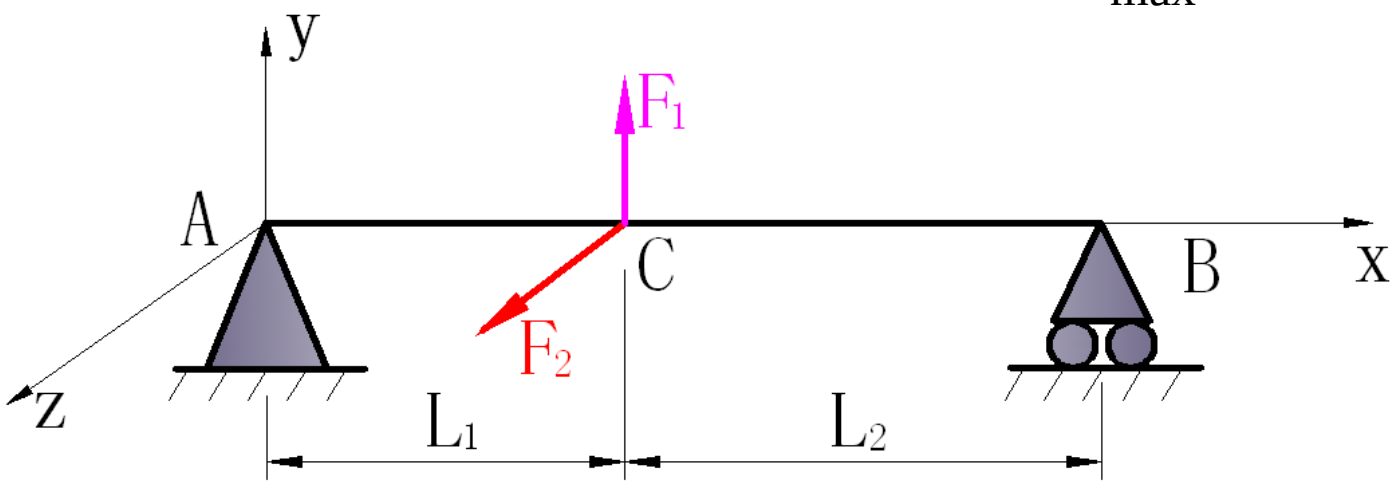
$y_c = 157.5mm$，$I_{zc} = 60130000mm^4$

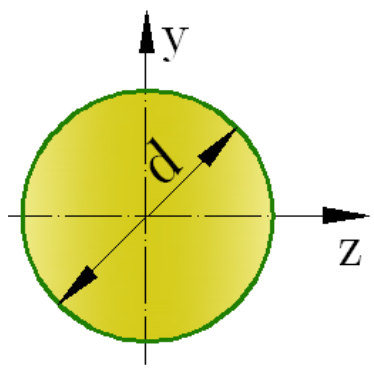
$\sigma_{Bt} = 24.1MPa$，$\sigma_{Bc} = 52.4MPa$，

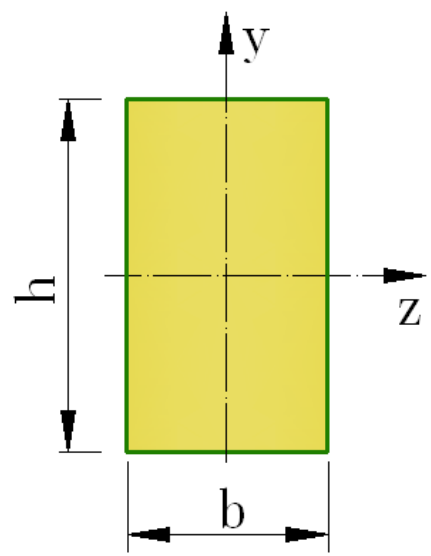
$\sigma_{Et} = 26.2MPa$

例题 5-4

如图简支梁，F_1作用在xy面内，F_2作用在xz面内，已知L_1、L_2。梁的截面有两种，一种是直径为d的圆形截面，一种是高为h，宽为b的矩形截面。求两种情况下全梁最大正应力σ_{max}。







圆形

$$\sigma_{max} = \frac{32 L_1 L_2 \sqrt{F_1^2 + F_1^2}}{\pi d^3 (L_1 + L_2)}$$

矩形

$$\sigma_{max} = \frac{6 L_1 L_2}{bh(L_1 + L_2)} \left(\frac{F_1}{h} + \frac{F_2}{b}\right)$$

§5-4 梁的切应力及强度条件

(Shear stresses in beams and strength condition)

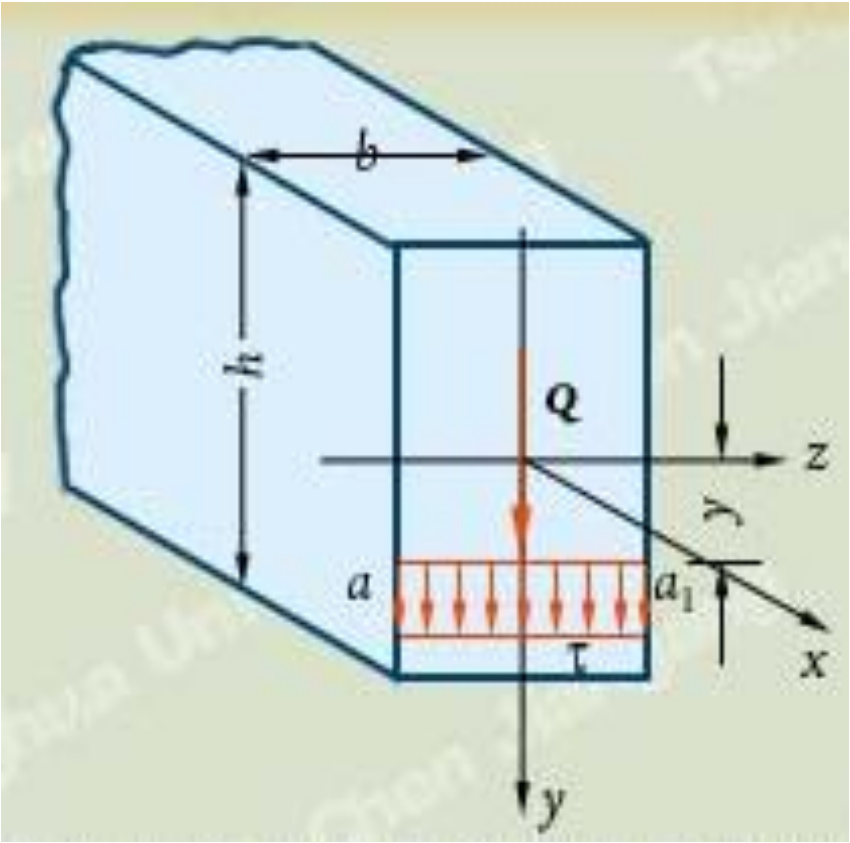
一、梁横截面上的切应力（Shear stresses in beams）

1、矩形截面梁(Beam of rectangular cross section)

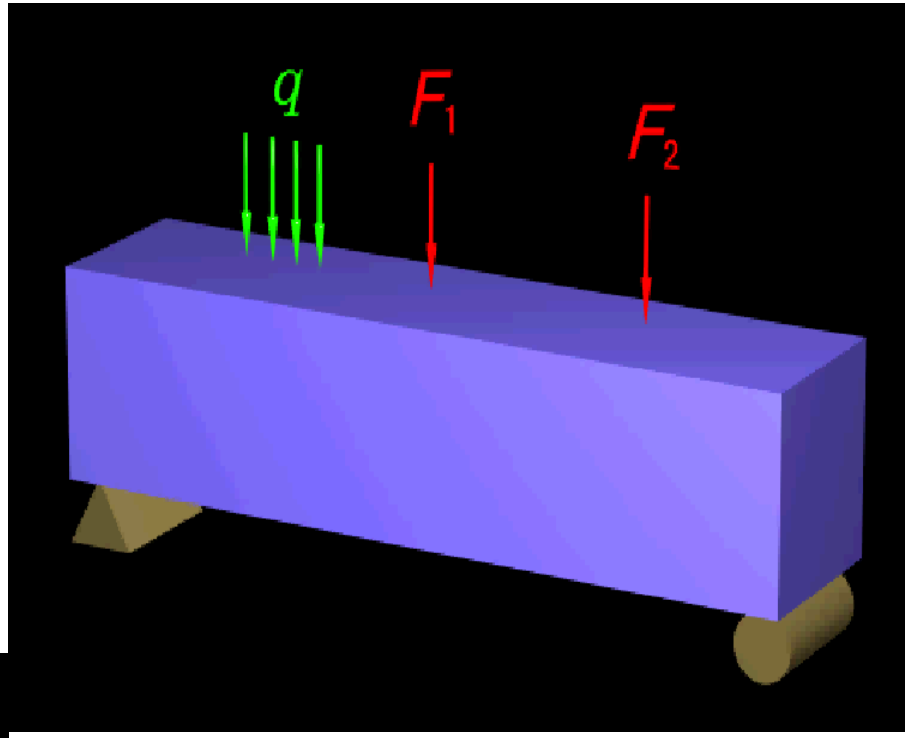
（1）两个假设(Two assumptions)

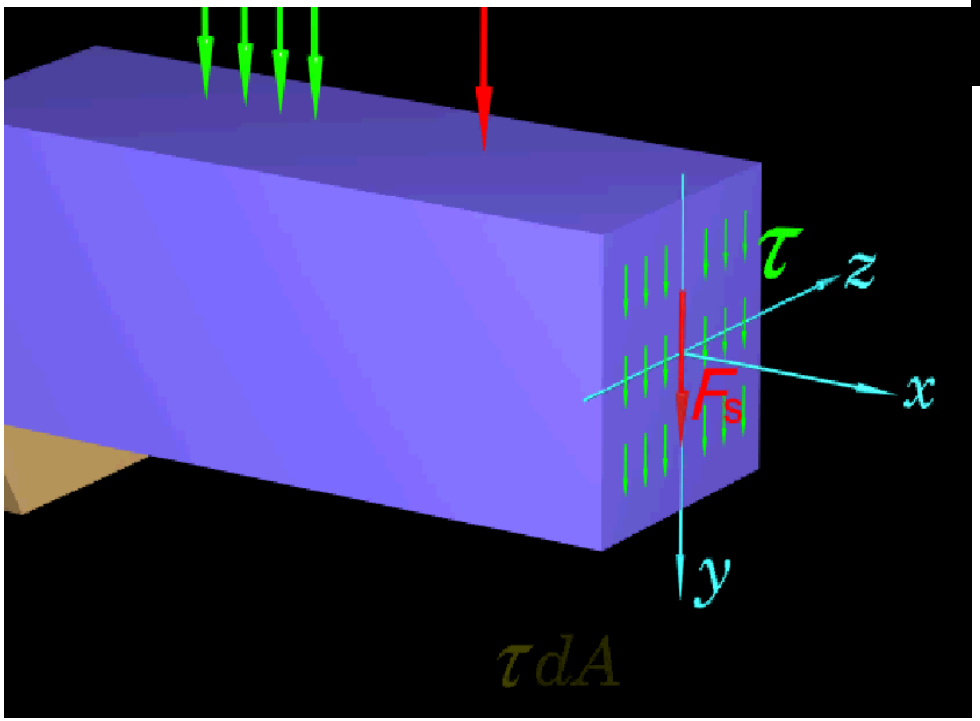
（a）切应力与剪力平行;

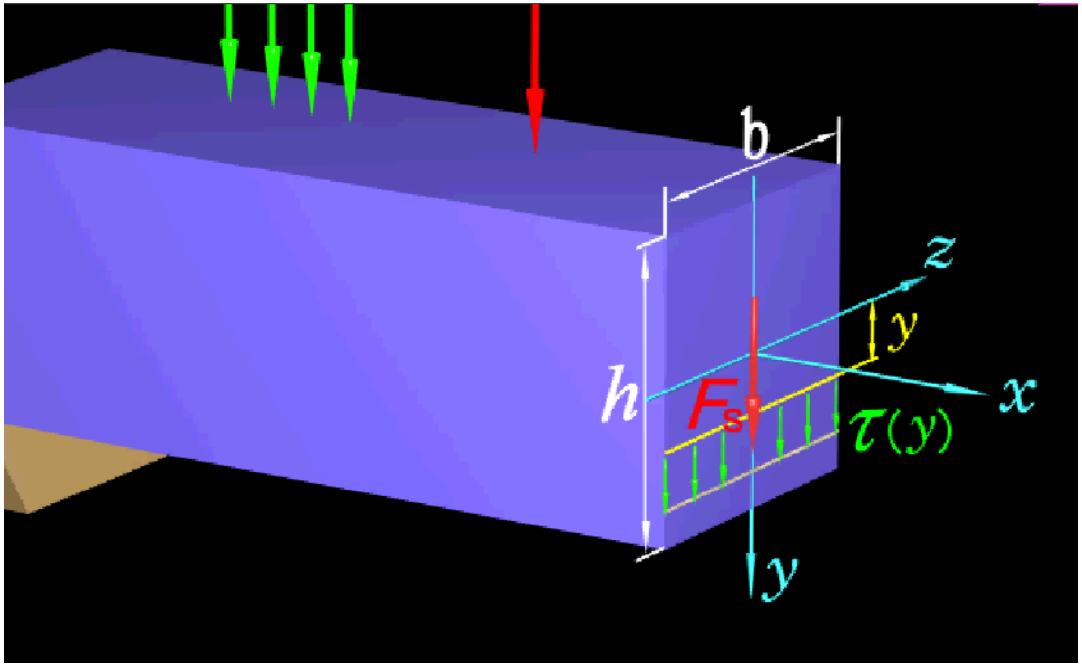
（b）切应力沿截面宽度均匀分布（距中性轴等距离处切应力相等）。

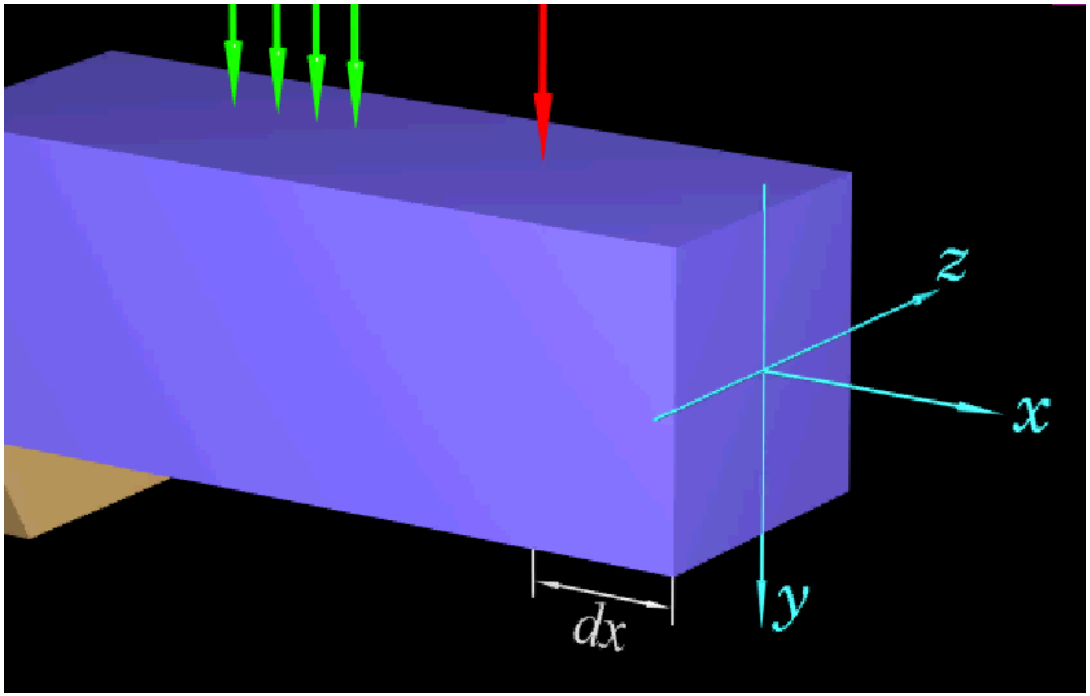


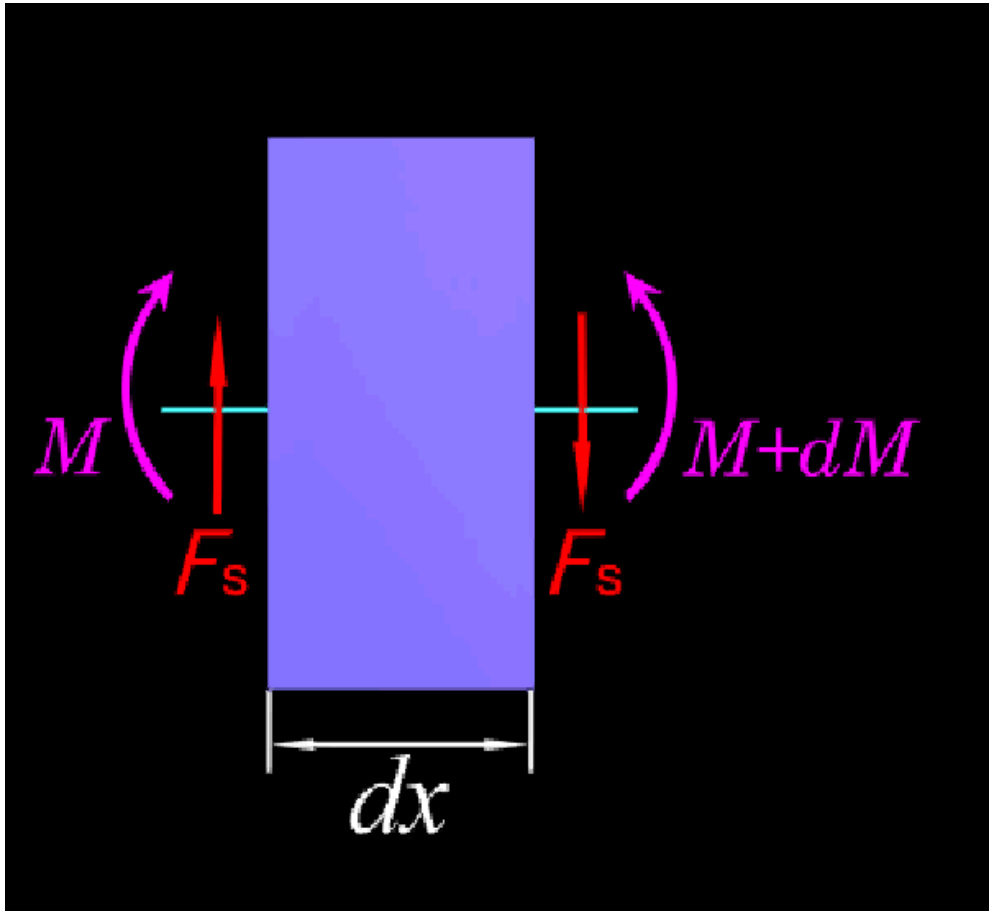
（2）公式推导

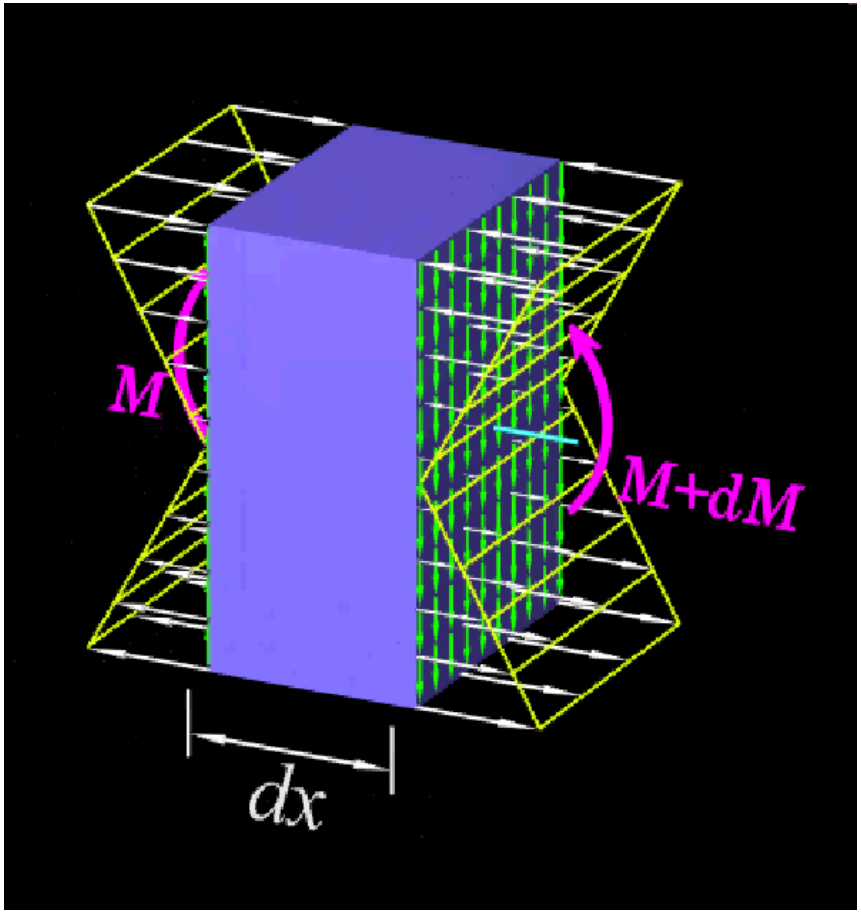


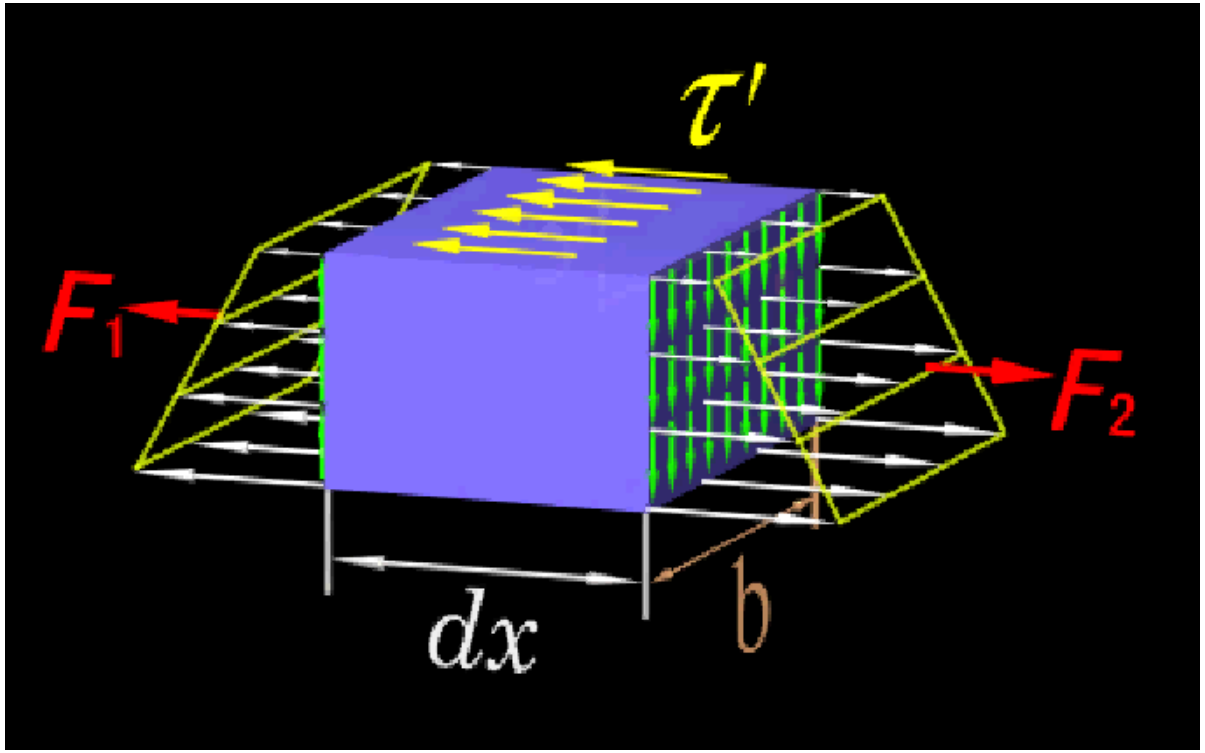


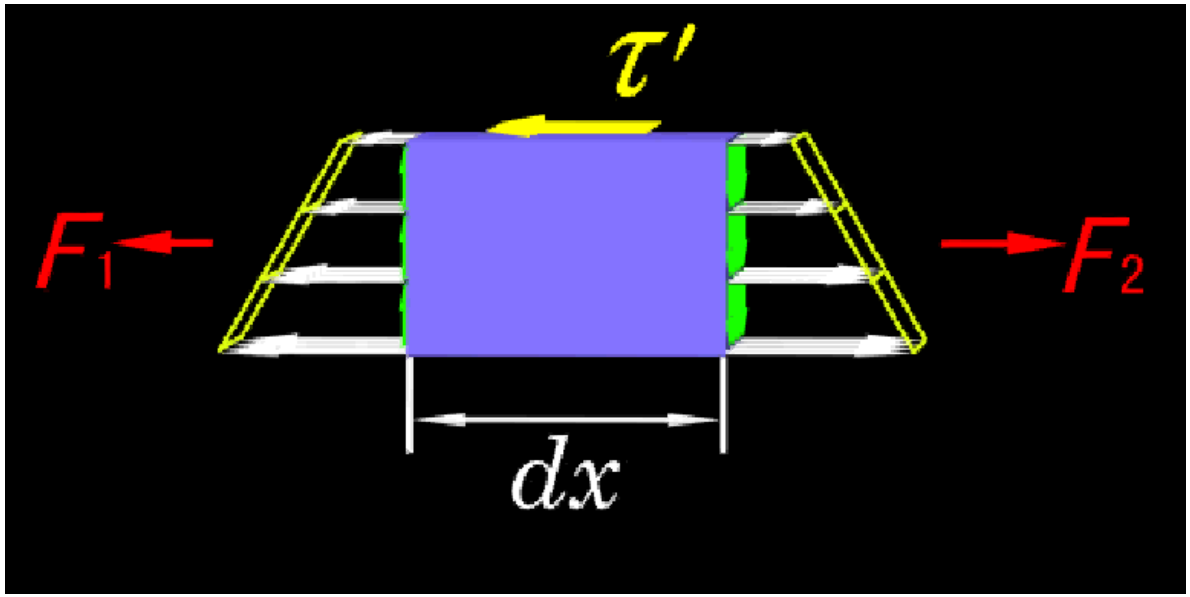

b
z
y
h
F_s
x
$\tau(y)$
y

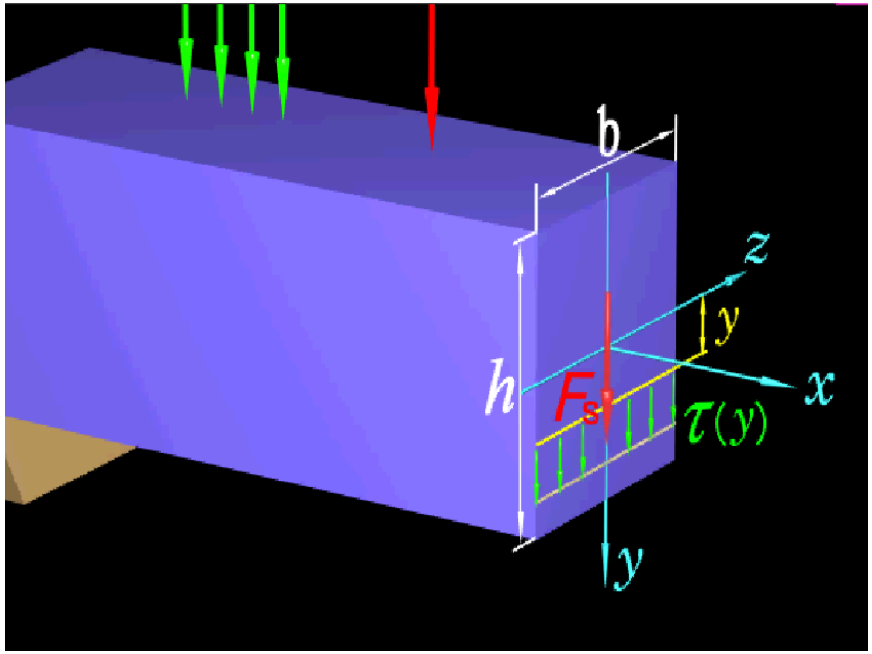

z
x
dx
y

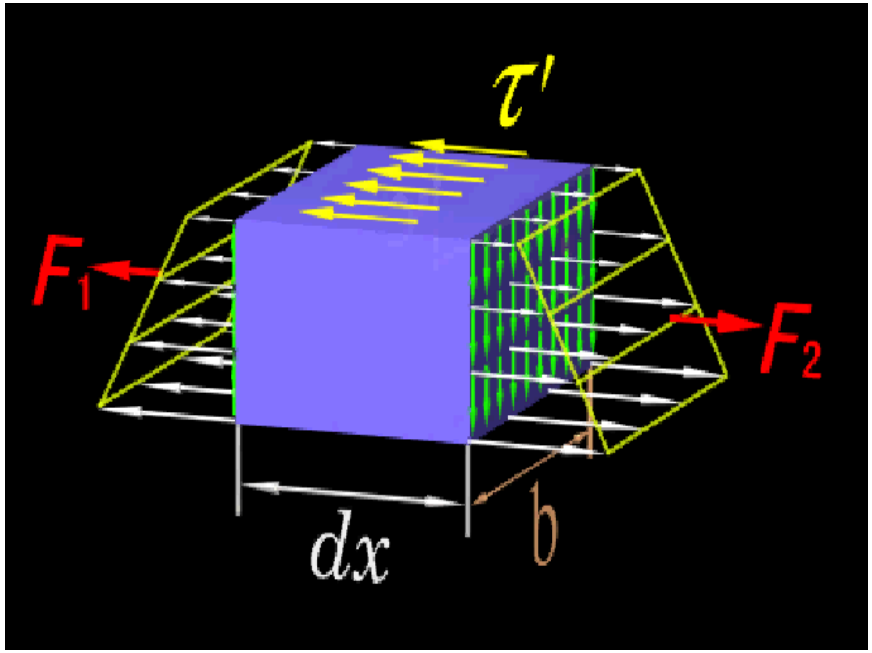

M
M+dM
Fs
Fs
dx


M
M+dM
dx


τ′
F1
F2
dx
b


τ′
F1
F2
dx





$$F_1 = \int_{A_1} \sigma_1 \mathrm{d}A = \int_{A_1} \frac{My_1}{I_z}\mathrm{d}A = \frac{M}{I_z}\int_{A_1} y_1 \mathrm{d}A = \frac{M}{I_z} S_z^*$$

$$F_2 = \int_{A_1} \sigma_2 \mathrm{d}A = \frac{M + \mathrm{d}M}{I_z} S_z^*$$

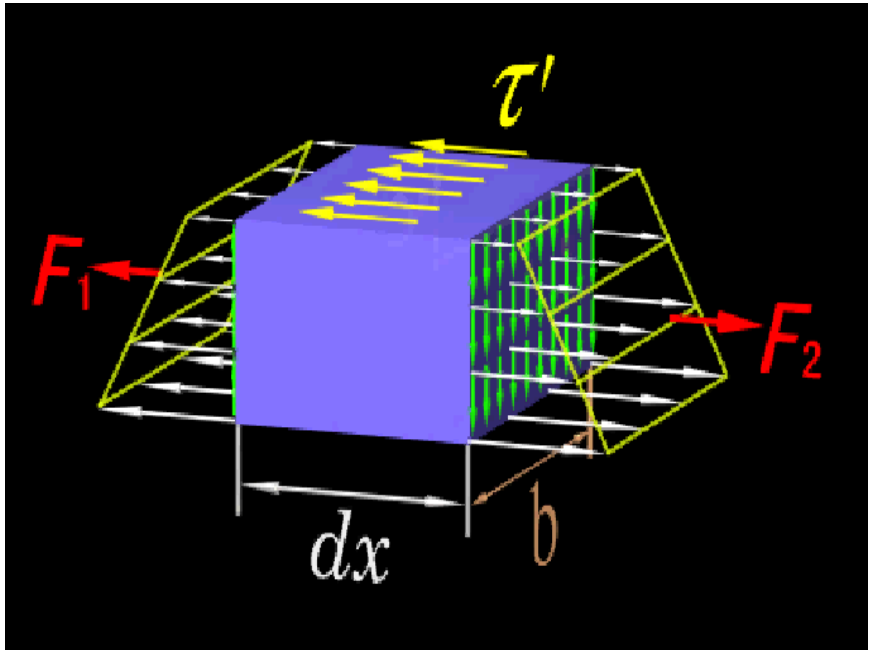
式中：A_1为距中性轴为y的横线以外部分的横截面面积。

$S_z^* = \int_{A_1} y_1 \mathrm{d}A$ 为面积A_1对中性轴的静矩。

$$F_1 = \frac{M}{I_z} S_z^*$$

$$F_2 = \frac{M + \mathrm{d}M}{I_z} S_z^*$$

$$\mathrm{d}F_s' = \tau' b \mathrm{d}x$$



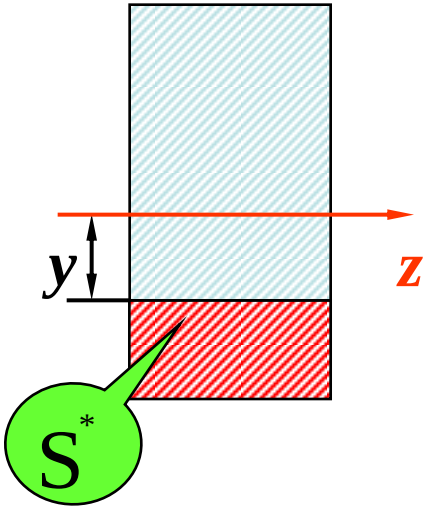
由平衡方程

$$\sum F_x = 0 \qquad F_2 - F_1 - \mathrm{d}F_S' = 0$$

化简后得

$$\tau' = \frac{\mathrm{d}M}{\mathrm{d}x} \times \frac{S_z^*}{I_z b} \qquad \frac{\mathrm{d}M}{\mathrm{d}x} = F_S \qquad \tau = \frac{F_S S_z^*}{I_z b}$$

$$\tau = \frac{F_S S_z^*}{I_z b}$$



I_z —— 整个横截面对中性轴的惯性矩。

b —— 矩形截面的宽度。

S_z^* —— 距中性轴为y的横线以外部分横截面面积对中性轴的静矩。

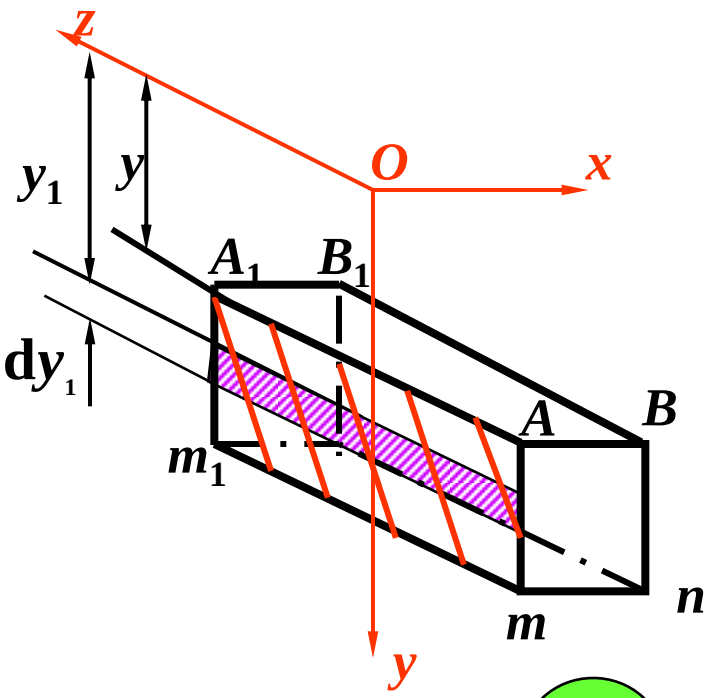
$$\boxed{\tau = \frac{F_S S_z^*}{I_z b}}$$

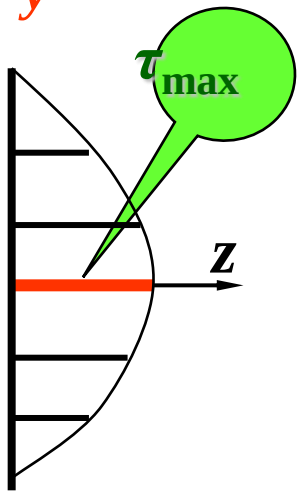
$$S_z^* = \int_{A_1} y_1 \mathrm{d}A$$

$$= \int_y^{h/2} y_1 b \mathrm{d}A_1 = \frac{b}{2}\left(\frac{h^2}{4} - y^2\right)$$

或 $S_z = \int_A y \mathrm{d}A = A\bar{y}$

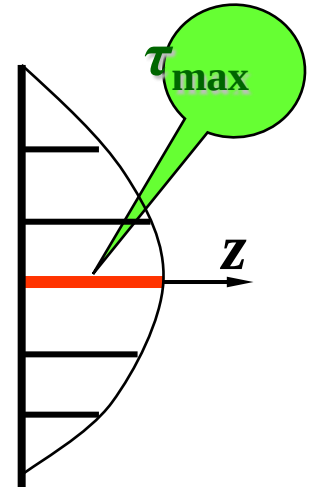
$$\tau = \frac{F_S S_z^*}{I_z b} = \frac{F_S}{2I_z}\left(\frac{h^2}{4} - y^2\right)$$





切应力沿截面高度按**抛物线规律**变化。

$$\tau = \frac{F_S S_z^*}{I_z b} = \frac{F_S}{2I_z}\left(\frac{h^2}{4} - y^2\right)$$



$y = \pm h/2$（即在横截面上距中性轴最远处）$\tau = 0$

$y = 0$（即在中性轴上各点处），切应力达到最大值

$$\tau_{max} = \frac{F_S h^2}{8I_z} = \frac{F_S h^2}{8 \times bh^3/12} = \frac{3}{2} \times \frac{F_S}{bh}$$

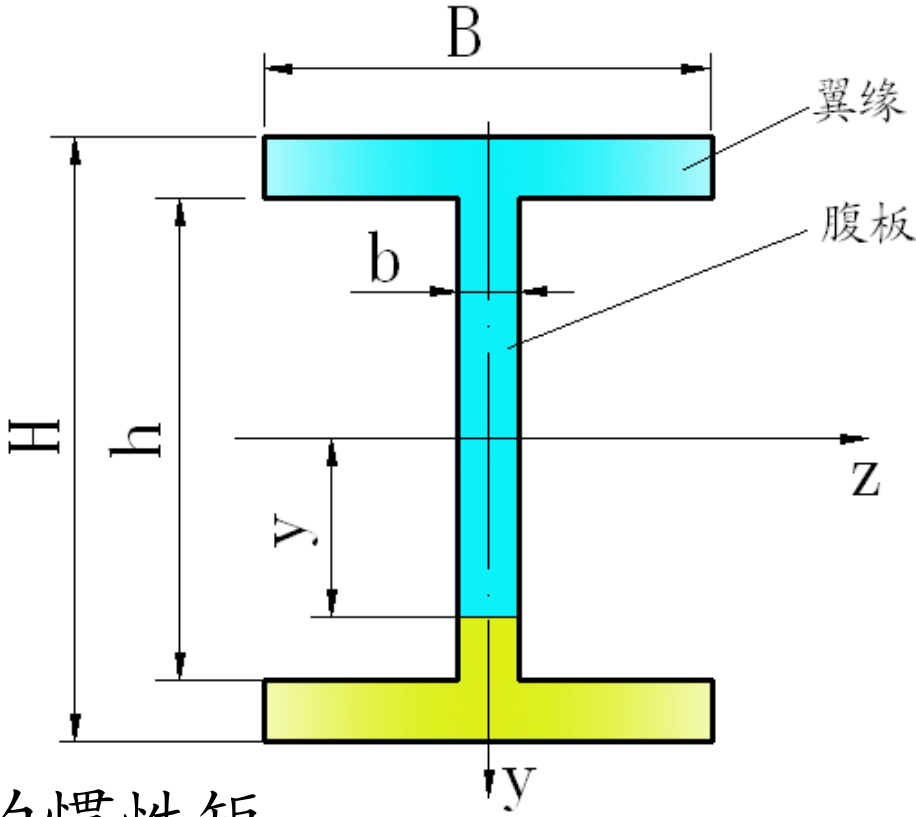
$$\tau_{max} = \frac{3F_S}{2A}$$

式中，A=bh为矩形截面的面积。

2、工字形截面梁（工-section beam)

研究方法与矩形截面同，切应力的计算公式亦为

$$\tau = \frac{F_S S_z^*}{I_z b}$$



假设求应力的点到中性轴的距离为y。

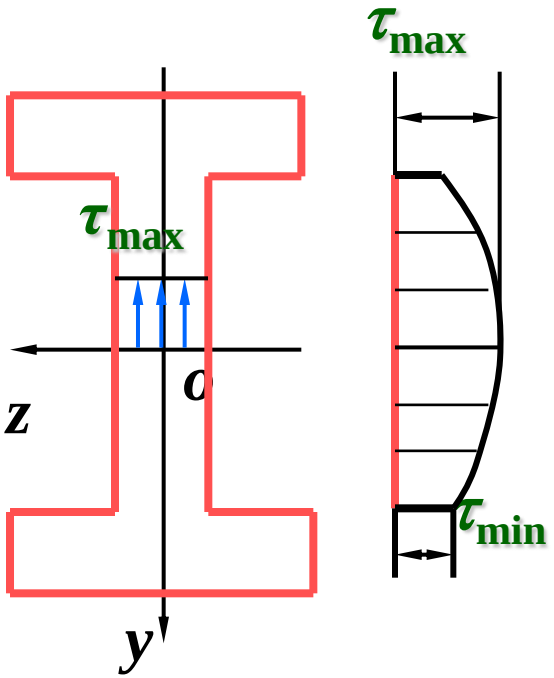
b —— 中性轴的宽度。

I_z —— 整个横截面对中性轴的惯性矩。

S_z^* —— 距中性轴为y的横线以外部分横截面面积对中性轴的静矩。

$$\tau_{\max}=\frac{F_S}{I_z b}\left[\frac{BH^2}{8}-(B-b)\frac{h^2}{8}\right] \qquad \tau_{min}=\frac{Q}{I_z b}\frac{B}{8}\left[H^2-h^2\right]$$

（a）腹板上的切应力沿腹板高度按二次抛物线规律变化。

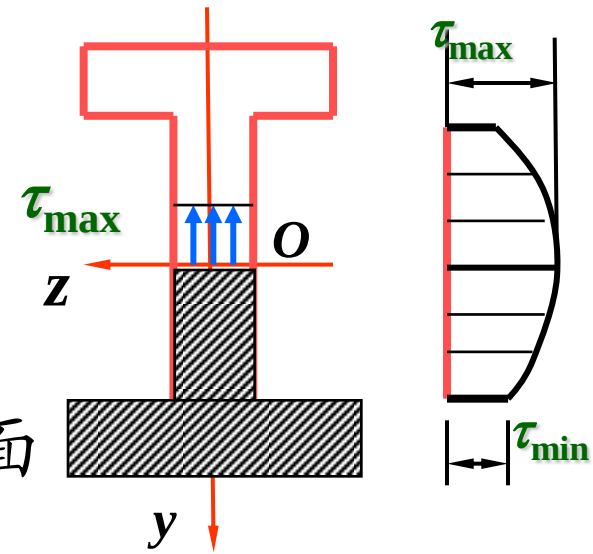


（b）腹板上最大切应力在中性轴上，这也是整个横截面上的最大切应力。

$$\tau_{\max}\approx\tau_{\min} \qquad \tau\approx\frac{F_S}{bh}$$

（c）翼板的切应力与腹板上切应力相比较小，工程上一般不考虑。腹板上能承担多少剪力？积分得总剪力的95％～97％。

$$\tau_{\max} = \frac{F_S S^*_{z\max}}{I_z b}$$



式中：

$S^*_{z\max}$——中性轴任一边的半个横截面面积对中性轴的静矩。

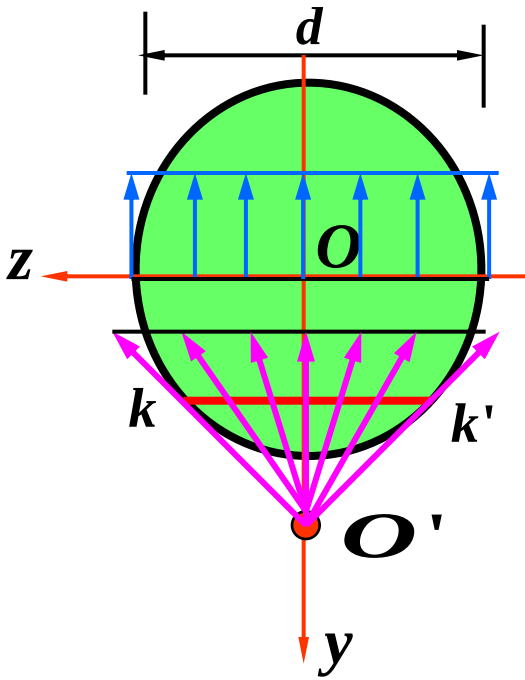
如为型钢，I_z、b、$S^*_{z\max}$查表。

3、圆截面梁(Beam of circular cross section)

最大切应力发生在中性轴上。

$$\tau_{\max}=\frac{F_S S_z^*}{I_z b}=\frac{4}{3}\frac{F_S}{A}$$

式中 $A=\dfrac{\pi d^2}{4}$ 为圆截面的面积。

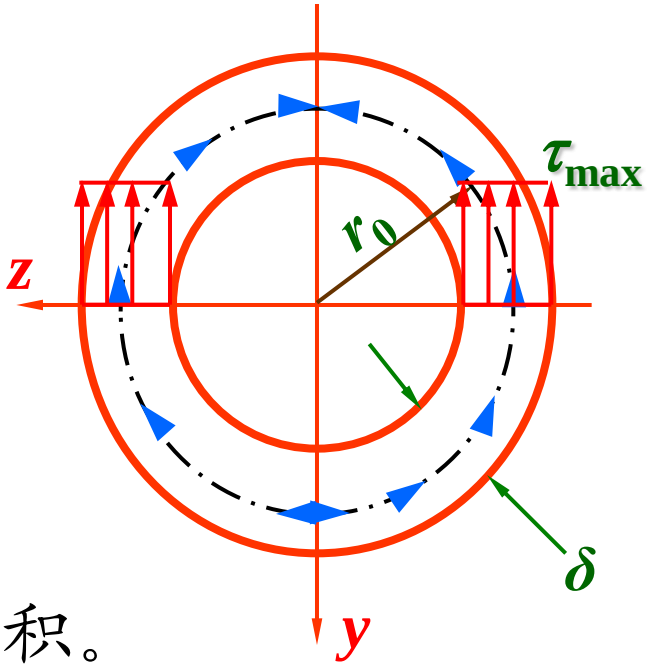


4、圆环形截面梁 (Circular pipe beam)

最大的切应力发生中性轴上。

$$\tau_{\max}=2\frac{F_S}{A}$$

式中 $A=2\pi r_0\delta$ 为环形截面的面积。



二、强度条件（Strength condition）

$$\tau_{\max} \le [\tau] \qquad \tau_{\max} = \frac{F_{S\max} S^*_{z\max}}{I_z b} \le [\tau]$$

三、需要校核切应力的几种特殊情况

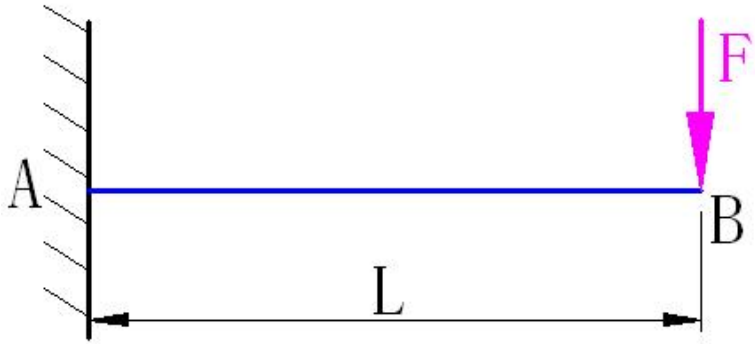
（1）梁的跨度较小，或支座附近有较大载荷，M 较小，而F_S较大时,要校核切应力;

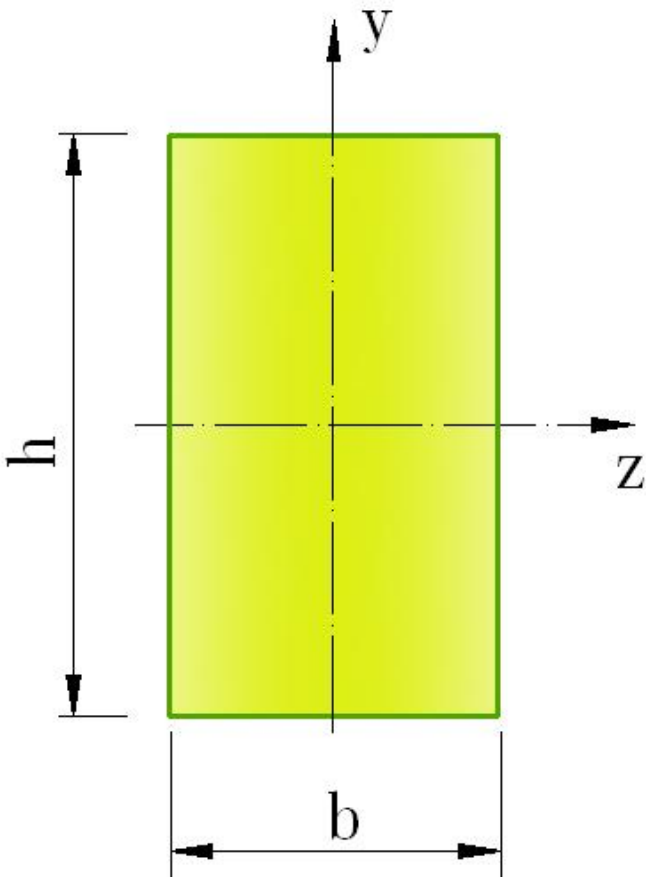
（2）铆接或焊接的组合截面，其腹板的厚度与高度比小于型钢的相应比值时，要校核切应力;

（3）焊接、铆接、胶合而成的梁，要对焊缝、胶合面等进行剪切强度计算

（4）各向异性材料(如木材)的抗剪能力较差，要校核切应力。

细长梁通常的控制因素是弯曲正应力，一般弯曲切应力都能满足。

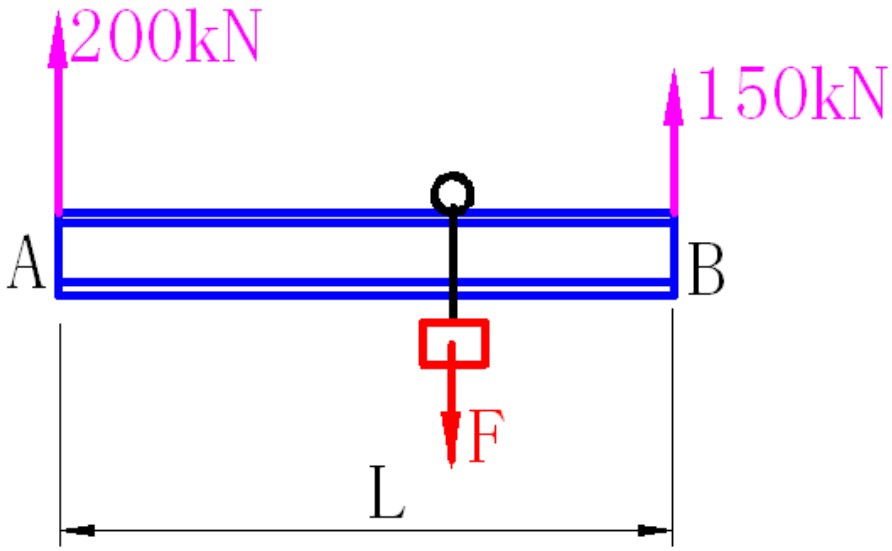


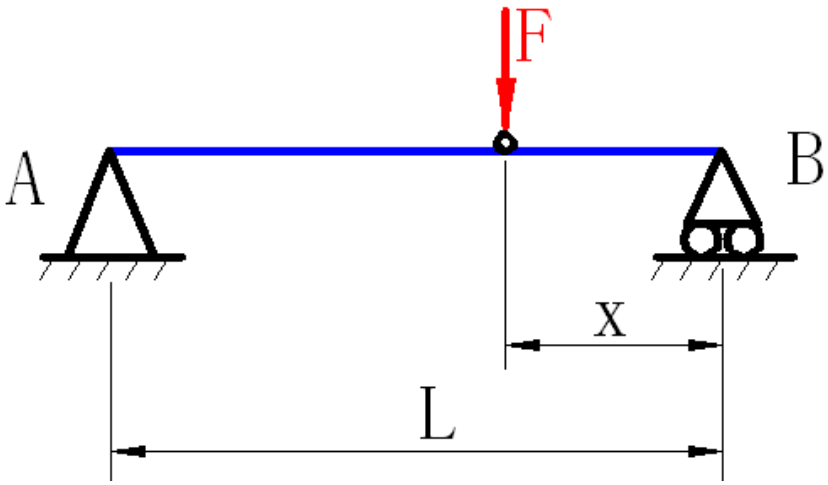


$$\sigma_{max} = \frac{6FL}{bh^2} \qquad \tau_{max} = \frac{3F}{2bh} \qquad \frac{\sigma_{max}}{\tau_{max}} = 4\frac{L}{h}$$

例题 5-6 为了起吊重F=300kN的大型设备，采用150kN和200kN的两台吊车，以及一根工字型辅助梁，组成临时附加悬挂系统，如图。辅助梁长L=4m，材料的许用正应力[σ]=160MPa，许用切应力[τ]=80MPa。

求：（1）设备加在辅助梁的什么位置才能保证两台吊车都不超载？（2）辅助梁的型号？





$$2\text{m} \le x \le \frac{8}{3}\text{m}，\ \text{NO.50a}$$

§5-5 提高梁强度的主要措施

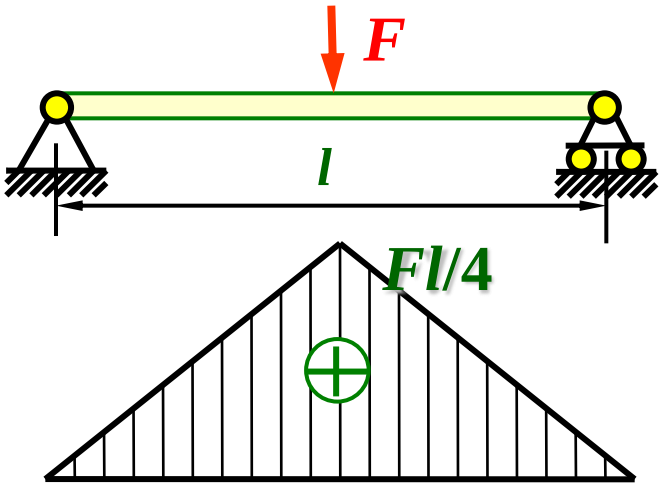
（Measures to strengthen the strength of beams)

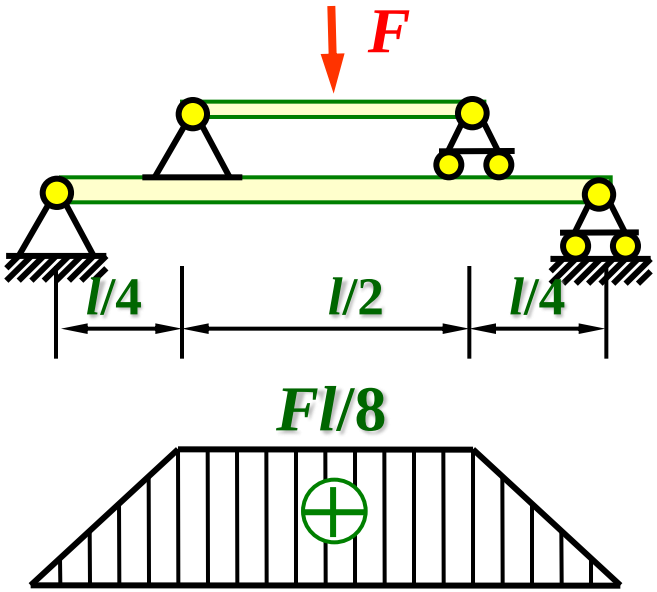
按强度要求设计梁时，主要是依据梁的正应力强度条件。

$$\sigma_{max} = \frac{M}{W} \le [\sigma]$$

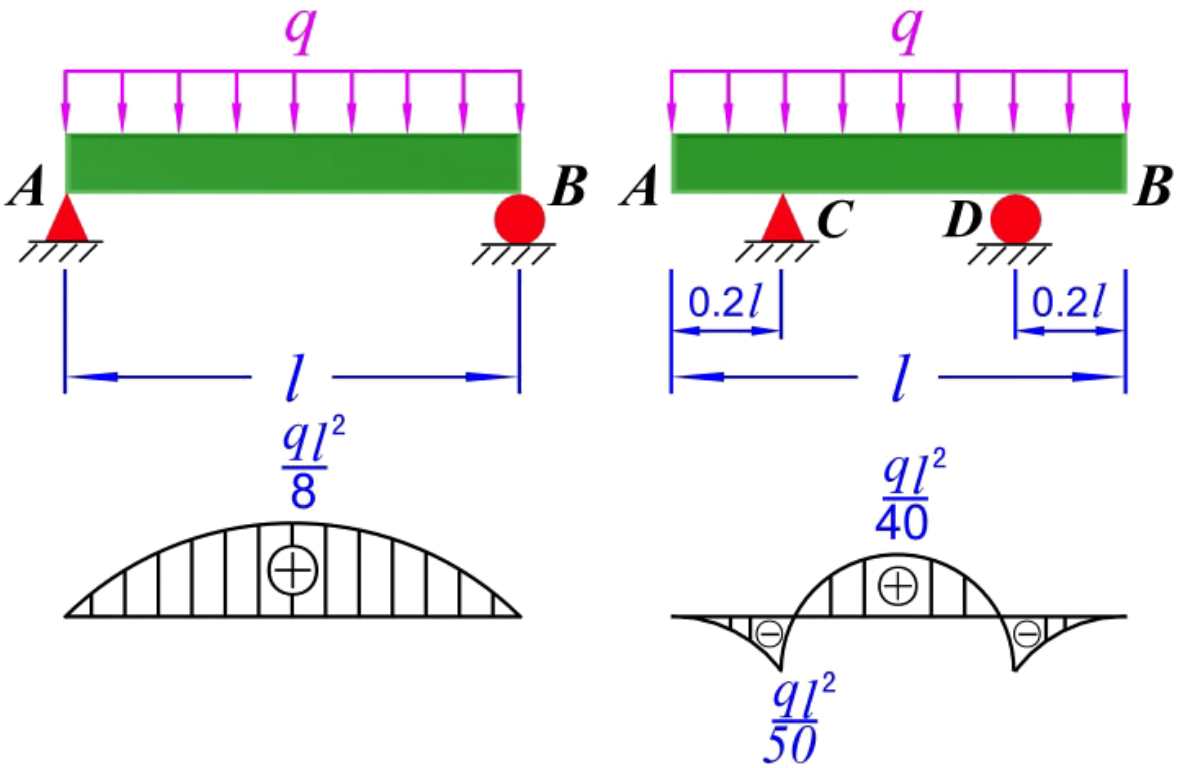
一、降低梁的最大弯矩值

1、合理地布置梁的荷载





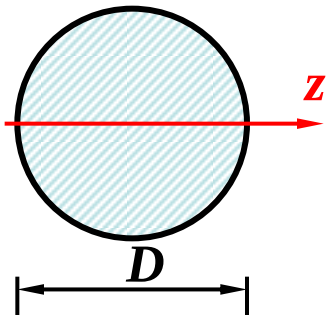
2、合理地设置支座位置



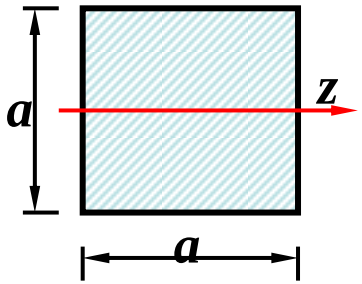
二、增大抗弯截面系数W

1、合理选择截面形状

在面积相等的情况下，选择抗弯模量大的截面。

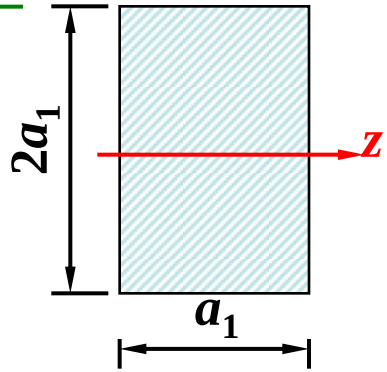


$$W_{z1} = \frac{\pi D^3}{32}$$



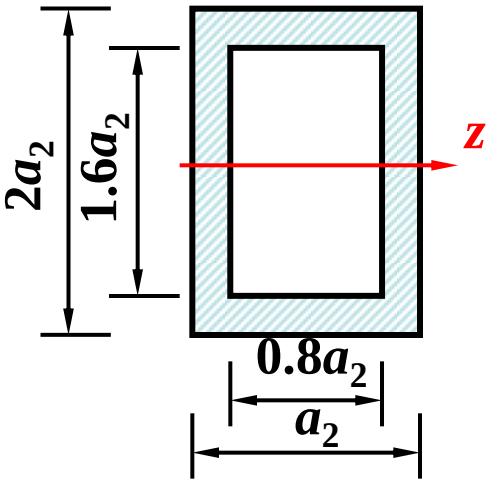
$$\frac{\pi D_1^2}{4} = a^2, a = \sqrt{\pi}(D_1/2)$$

$$W_{z2} = \frac{bh^2}{6} = \frac{(\sqrt{\pi}R)^3}{6} = 1.18W_{z1}$$



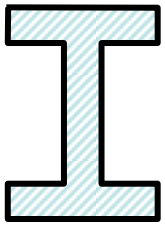
$$\frac{\pi D_1^2}{4} = 2a_1^2, a_1 = \sqrt{2\pi}D_1$$

$$W_{z3} = \frac{bh^2}{6} = \frac{4a_1^3}{6} = 1.67W_{z1}$$



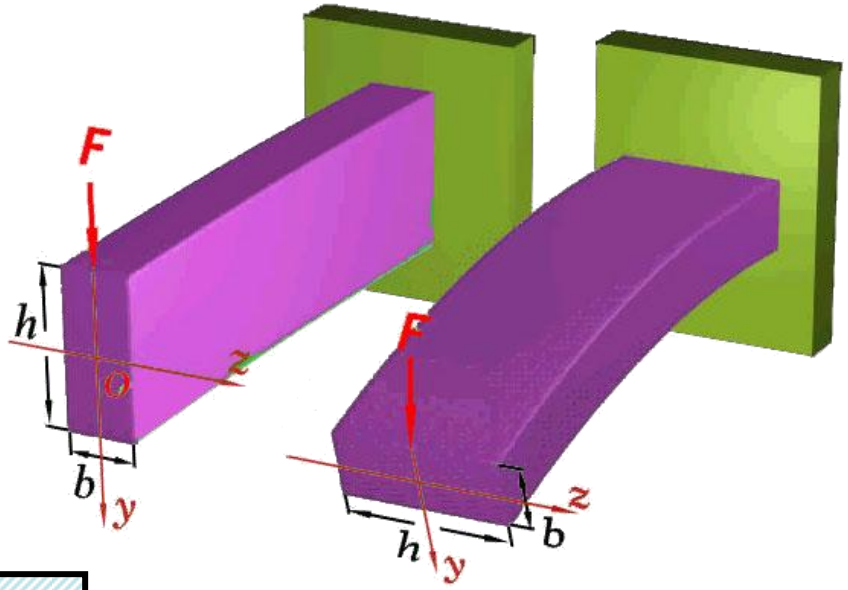
$$\frac{\pi D_1^2}{4} = 2a_2^2 - 0.8 \times 1.6a_2^2, a_2 = 1.05D_1$$

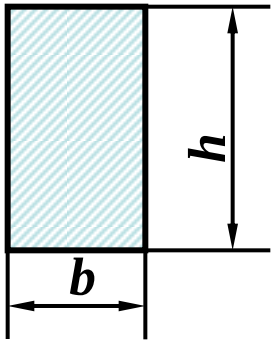
$$W_{z4} = 4.57W_{z1}$$

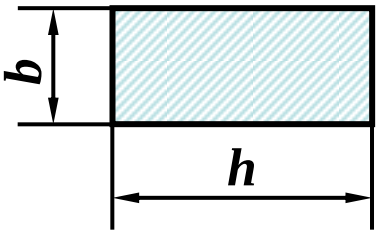


工字形截面与框形截面类似。

2、合理的放置







$$W_1 = \frac{bh^2}{6}$$

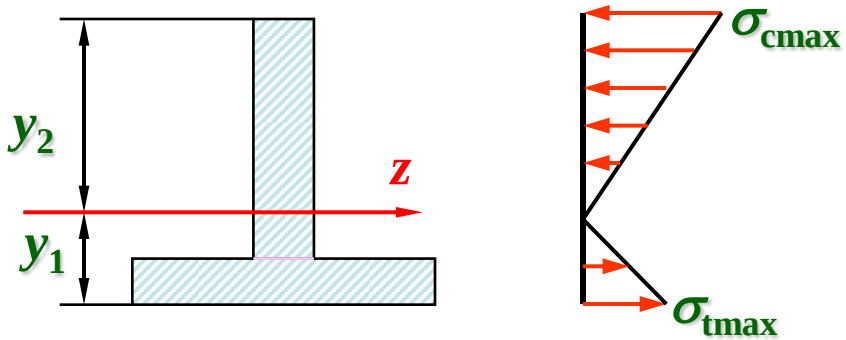
$$W_2 = \frac{hb^2}{6}$$

$$\frac{W_1}{W_2} = \frac{h}{b}$$

三、根据材料特性选择截面形状

1、对于塑性材料制成的梁，选以中性轴为对称轴的横截面

2、对于脆性材料制成的梁，宜采用T字形等对中性轴不对称的截面且将翼缘置于受拉侧。



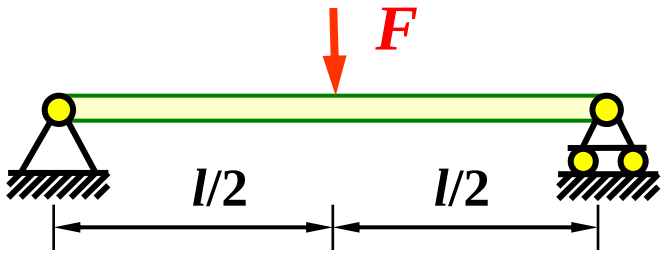
要使y_1/y_2接近下列关系：最大拉应力和最大压应力同时接近许用应力。

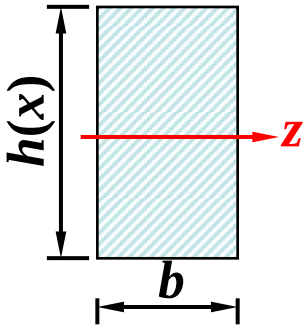
$$\frac{\sigma_{tmax}}{\sigma_{cmax}} = \frac{\dfrac{M_{max} y_1}{I_z}}{\dfrac{M_{max} y_2}{I_z}} = \frac{y_1}{y_2} = \frac{[\sigma_t]}{[\sigma_c]}$$

四、采用等强度梁

梁各横截面上的最大正应力都相等，并均达到材料的许用应力，则称为等强度梁。

例如，宽度b保持不变而高度可变化的矩形截面简支梁，其高度随截面位置的变化规律h(x)，可按正应力强度条件求得。

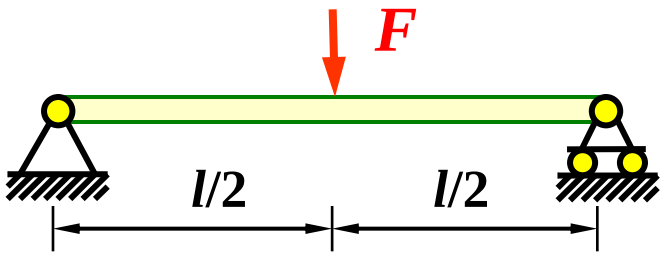


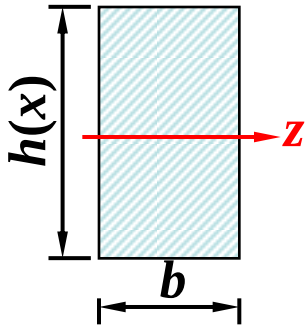


梁任一横截面上最大正应力为

$$\sigma_{\max}=\frac{M(x)}{W(x)}=\frac{(F/2)x}{(1/6)bh^2(x)}\le[\sigma]$$

求得 $h(x)=\sqrt{\dfrac{3Fx}{b[\sigma]}}$



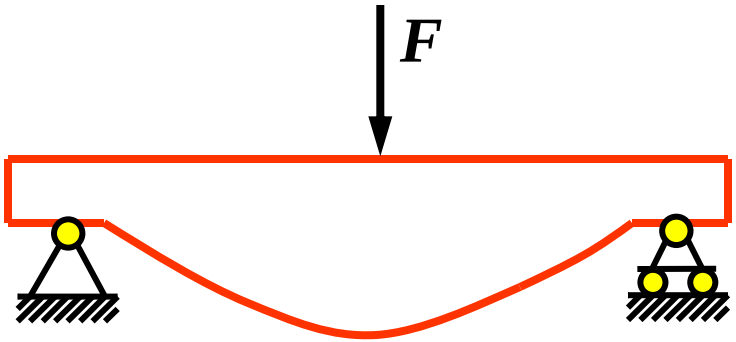


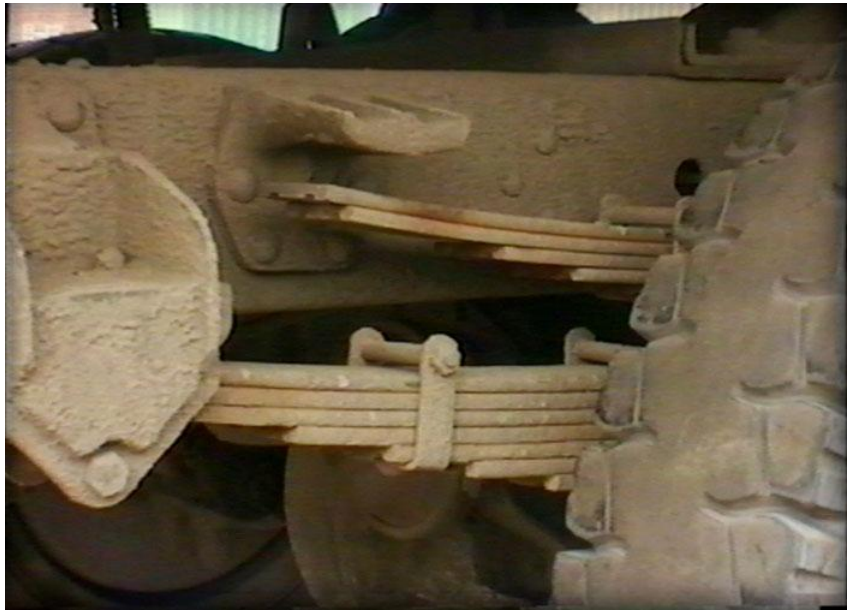
$$h(x)=\sqrt{\frac{3Fx}{b[\sigma]}}$$

但靠近支座处，应按切应力强度条件确定截面的最小高度。

$$\tau_{\max}=\frac{3}{2}\frac{F_S}{A}=\frac{3}{2}\frac{F/2}{bh_{\min}}=[\tau]$$ 求得 $$h_{\min}=\frac{3F}{4b[\tau]}$$

按上确定的梁的外形，就是厂房建筑中常用的鱼腹梁。





第五章结束